

Progress on the demand side management in smart grid and optimization approaches

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Summary

The integration of demand side management (DSM) with smart grid (SG) can facilitate residents' transfer into smart homes and sustainable cities by reducing the carbon emission. This manuscript reviews the recent works related to the application of DSM in SG through discussing the techniques and algorithms and their associated challenges for effective implementation. This paper also critically discusses the operation mode of DSM, the profile of energy production, storage and consumption, and finally the benefit obtained by the DSM implementation. Previous literature suggested that DSM practice reduced peak-to-average ratio, energy cost and carbon emission by approximately 10% to 65%, 5% to 50%, and 14%, respectively. The implementation of DSM in SG deals with a number of challenges such as security and privacy, tariff regulation, energy transmission, distribution, and effective utilization of energy resources. A number of international organizations have taken various measures and solutions to guarantee the security and privacy of the DSM in SG discussed. So far, a number of algorithms have been used as optimization approach to solve the DSM optimization problems; however hybrid algorithms have showed better performance than single algorithms due to their faster convergence speed. At the end, the paper presents the research gaps and future research directions.

KEYWORDS

algorithms, demand side management, load management, optimization, peak loads, smart grid, sustainability

Abbreviations: ACO, ant colony optimization; AMI, advanced metering infrastructure; BA, bat algorithm; BCSA, bat-crow search algorithm; BFO, bacterial foraging optimization; BPSO, binary particle swarm optimization; CPP, critical peak pricing; CSA, crow search algorithm; DLC, direct load control; DES, distributed energy storage; DP, dynamic programming; DR, demand response; DSM, demand side management; EDE, enhanced differential evolution; EE, energy efficiency; ESS, energy storage system; EU, Euclidian; GA, genetic algorithm; GTA, game theory algorithm; GWO, gray wolf optimization; HEMS, home energy management systems; HGPSO, hybrid algorithm consisting of GA and PSO; HGWD, hybrid genetic wind-driven; HS, harmony search; IPSO, improved PSO algorithm; LP, linear programming; MINLP, mixed-integer nonlinear programming; NLP, non-linear programming; PAR, peak-to-average ratio; PSO, particle swarm optimization; RES, renewable energy resources; SG, smart grid; SR, spinning reserve; TLBO, teacher learning-based optimization; TLGO, teacher learning genetic optimization; TOU, time of use; VCG, Vickrey-Clarke-Groov; WDO, wind-driven optimization.

1 | INTRODUCTION

Energy is an essential element of people's daily life. At present, social development depends on the usage of a sufficient amount of energy, especially electricity.^{1,2} The amount of energy consumption increases along with the rapid growth in global population and industrial and technological development.³ Global energy consumption primarily depends on the fossil fuel resources such as oil, natural gas, and coal.⁴⁻⁶ Thus, it is quite difficult to meet the future energy demand due to the limited fossil fuels. In addition, the fossil fuels are responsible for the high carbon emission and global warming. According to the International Energy Agency, approximately 70% of world's total energy is produced through the burning of fossil fuels, primarily coal (42%) and gas (21%).⁷ Currently, maintaining an adequate electricity supply is considered one of the most challenging tasks for ensuring continuous economic and industrial development.

Environmental sustainability and energy safety are associated with the production and consumption of energy. A large amount of energy resources are wasted through the unproductive use of natural resources. Moreover, some countries, such as the United States, use natural gas and coal-based power plants to produce electricity.⁵ These fossil fuel-based power plants produce a large amount of SO₂, CO₂, and other greenhouse gasses, all of which threaten the environment. The scientific community has begun to search the alternative energy options for power generation because of the growing demand of electricity and the contribution of power plants to climate change.⁸

Electric network performance depends on the balance between the production of electricity and the capability to meet consumers' growing demand. In addition, the amount of energy consumption certainly affects the energy distribution system. This phenomenon makes the operation of the grid unsafe and unreliable. Therefore, the variability in renewable energy generation needs to be considered to meet the growing power demand and ensure grid sustainability. The smart grid (SG) is an electric network with smart meters, advanced sensing technologies, control systems, and communication technologies that reflect the future of energy systems.^{9,10} The theory behind the SG has been evolved with the proper and effective distribution and supply of electricity and management of power load. The main characteristics of SG includes the bidirectional flow of data and energy between the energy provider and the customer.¹¹ Therefore, the SG opens the door for new prospects to supply electricity to the consumer efficiently and dynamically.

The issues associated with grid sustainability and reliability can be addressed by demand side management

(DSM). The DSM is a set of load management actions, which includes planning, implementation, and monitoring of pre-defined activities that affect the consumers' electricity utilization patterns.¹² The DSM can systematically transmit and distribute available energy to decrease carbon emissions and peak loads as well as allows users to choose their preferred energy type.¹³ The DSM was first devised in the year 1970.¹⁴ The DSM model was invented by the electricity industry to control the time of use (TOU) and level of electricity demand and to analyze the profiles of electricity loads among users. A DSM program integrated with renewable energy sources (RES, for example, solar and wind), distributed micro-generators and energy storage devices, such as plug-in electric vehicles and batteries can provide an optimal management system by scheduling various smart appliances and generating renewable energy.¹⁵⁻¹⁷ The price of electricity significantly affects the usage of energy by the consumers.^{18,19} However, the implementation of the DSM in SG can easily handle the load patterns of the electricity market as well as can analyze and reshape load profiles. This practice reduces the peak load demand of customers, thereby improving grid stability, sustainability, and security; additionally reduces carbon emission levels, grid operation costs, and electricity costs.²⁰ Also, effective DSM activities can easily avoid the unnecessary construction of electrical infrastructure by controlling and managing decentralized energy resources, which includes controllable end use devices. These activities can manage the electricity market by considering power generation, transmission, and distribution.

The algorithms can solve the optimization problems of DSM in SG. These optimization problems include minimization of electricity costs, aggregated power consumption, and peak-to-average ratio (PAR) as well as maximization of user comfort and the efficient integration of RESs.²¹ For example, previous studies presented different genetic algorithm (GA)-based optimization models for reducing electricity costs.²²⁻²⁴ Additionally, GA, ant colony optimization (ACO) and binary particle swarm optimization (BPSO) were used to schedule energy consumption and evaluate the performance of home energy management systems (HEMS) controllers.²⁵

Several review papers have been published on the DSM of SG mostly focused on the distributed generation with renewable energy integration, optimal load scheduling of demand response (DR), and innovative enabling technologies and systems.^{26,27} However, the current review paper is novel in the sense that the paper critically discusses the available optimization approaches for DSM, challenges of DSM implementation in SG and their associated solutions, which have not been explored in the previous review paper. The paper also revisits and

scrutinizes the strategies of DSM and the role of distributed energy generation from renewables and storage systems on SG. This state-of-the-art review is structured as follows. Section 2 presents the impacts of DSM practice for achieving the sustainability of SG. Section 3 and Section 4 describes the techniques of DSM and challenges of DSM with their possible solutions, respectively. The role of distributed generation from renewable resources in SG is presented in Section 5. Section 6 deals with the comprehensive review on the insight of the DSM optimization approaches followed by comparison in Section 7. Finally, the review provides future research directions in the conclusion section.

2 | SIGNIFICANCE OF DSM ON SG SUSTAINABILITY

The SG is considered a new opportunity to enhance the 21st century's power grid. The SG has gained substantial popularity due to certain features, such as distributed generation, self-healing, digital two-way communication, self-monitoring, and universal control.^{28,29} The SG can adjust renewable energy generation, create smart measurement systems, and distribute and transmit grid power by utilizing modern information and communication technologies.^{30,31} In addition, the SG can control and manage the electricity market, construct the infrastructure, and manage the decentralized energy resources.³² The DSM supports the SG functionalities by analyzing the short and long term status of the electricity market, determining a cost-effective option for energy supply, and modeling and characterizing the system load.³³ However, the capacity of SG needs to be improved to meet the growing energy demand, which requires an installation of power generation and transmission infrastructure.³⁴ The development of new infrastructure will increase not only the complexity of the SG networks but also relevant system costs. In this situation, the efficient implementation of DSM programs in SG can overcome complexity and high expenses by controlling and influencing energy demand. Additionally, the DSM can improve grid sustainability by reducing the peak load demand, reshaping load profiles, and reducing overall costs and carbon emissions. Previous studies reported the contributions of DSM on the reduction of carbon emissions of SG. For instance, Zhang et al reduced the carbon emission levels in the SG environment by incorporating electric vehicles and a DR program.³⁵ Ai et al introduced a bid-scheduling DSM method in SG to motivate consumers leading to the reduction of carbon emission levels.³⁶ In this case, they developed demand side reserve scheduling based on the DR program in the SG environment. Li et al proposed a carbon emission flow

model to investigate and evaluate carbon emission levels from different parts of an SG network.³⁷ DSM and supply-side management jointly take part in the energy management program. The effects of incorporating three levels of RES utilization were also investigated for the reduction of carbon emission.

3 | TECHNIQUES AND APPROACHES OF DSM

The effective implementation of DSM changes the time of load consumption and the utility's total load, thereby reduce the expected peak loads.³⁸ In order to reduce electricity costs, the DSM manipulates customers' electricity usage patterns and produces the preferred changes in the load profiles by altering the load shape of the power distribution network.^{39,40} Essentially, the DSM helps to avoid excess power generation by reducing peak loads leading to a reduction in operating costs and capital expenditures.⁴¹ The DSM can be categorized into six major types according to the daily and seasonal usage of electricity (Figure 1).¹ These methods include load shifting, peak clipping, strategic load growth, valley filling, strategic conservation, and flexible load shape.⁴³⁻⁴⁵

- **Peak clipping:** Peak clipping is a common form of the load management technique that decreases the peak demand of an electrical network.³⁸ Typically, peak clipping controls customers' electricity consumption through direct load control (DLC), which mainly explains the system's peak load reduction during specific periods of time.^{38,40} The DLC can be defined as a function of the DSM program by which a power supplier company regulates customers' appliances from a distance.^{46,47} The proper scheduling of DLC is considered as a favorable way to reduce operating costs and fossil fuel usage.^{38,48}
- **Valley filling:** Valley filling aims to drop the level of load difference between the peak load and the valley

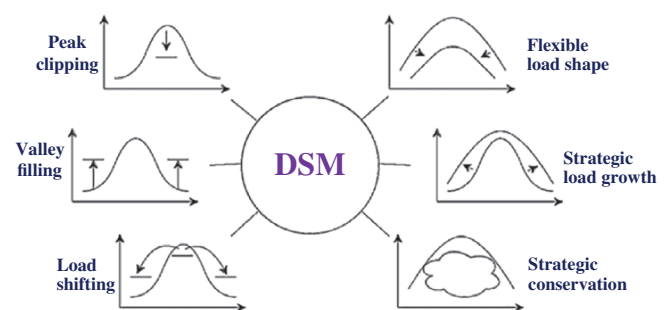


FIGURE 1 demand side management load shaping techniques⁴² [Colour figure can be viewed at wileyonlinelibrary.com]

load and thereby diminish load demand by filling the valley from a curtailed load.^{40,43} Valley filling is applicable when the long-run incremental cost is less than the average price of electricity.⁴³

- **Load shifting:** The load shifting technique is mostly used in the DSM program. It is the most effective load management technique that shifts load from peak hours to off-peak hours by the combined effects of peak clipping and valley filling.^{40,43}
- **Strategic conservation:** Strategic conservation diminishes overall load demand through the application of load reduction procedures by the efficient consumption of energy. It designs and attains the desired load shape according to the planning, distribution, and management of the network system.^{40,44}
- **Strategic load growth:** Strategic load growth motivates power companies to increase the power generation for customers.^{40,43,49} It optimizes the daily response and changes the shape of the load with respect to the large demand beyond the valley filling technique. The activities of strategic load growth include the amplification of the market share of loads, the economic development of service areas, and the guaranteeing of necessary infrastructure for handling the load demand.
- **Flexible load shape:** The flexible load shape technique mainly secures the reliability of SG.^{40,43,49} Under this technique, an electricity generation company analyzes the load profile to identify customers with flexible loads. The customers have been rewarded various incentives, if they control their consumption of interruptible or curtailable load during peak periods.

Recently, governments and utility companies have focused on the implementation of the DSM strategies that smooth the operation of electrical systems,⁵⁰⁻⁵² promote and extend energy efficiency (EE) plans and applications, and change the behavior at the customer level or implement dynamic DRs.^{53,54} The DSM has four strategies, namely, DR, EE, spinning reserve (SR), and TOU.⁵⁵⁻⁵⁷ The main emphasis of these strategies lies in the development and use of power-saving technologies, monetary incentives, electricity price, and government policies to diminish peak load demands and maintain a sophisticated synchronization between network operators and customers.

- **Energy efficiency:** The EE is considered a modest choice with respect to the benefits received by energy suppliers, energy consumers, and the environment.⁵⁸ The EE is a type of technology that provides an improved and long-lasting service when the end-use equipment is in operation. The EE programs improve the physical infrastructure of electrical grid for improving the electricity efficiency and reducing peak demand.⁵⁹

- **Time of use:** A TOU pricing strategy refers to a function of fixed tariff that divides 24 hours into several time intervals and assigns a different price for electricity consumption in each interval.^{59,60} This strategy helps to control peak period pricing and seasonal pricing based on different prices of energy.
- **Demand response:** The DR refers to a specific tariff or program, which intended to decrease or shift the electricity usage during peak periods with respect to time-based rates or incentive payment programs.^{61,62} The network reliability of an electrical system becomes jeopardized due to certain conditions, such as peak period network congestion or high prices. In this situation, the DR changes the energy usage pattern and provides an opportunity to consumers to contribute in the operation of the electric grid.⁵³ The DR programs can be classified into price-based programs and incentive-based programs, as illustrated in Figure 2.
- **Spinning reserve:** The SR refers the reserve power connected to the grid system, which is activated by the system operator to maintain the balance between load and generation in the case of a sudden drop in generation.⁶⁴ The interruption in power supply is caused due to an unexpected damage in generation units, incorrect load forecasting, and scheduling.⁶⁵ Typically, the SR is classified into primary and secondary SR.⁵⁹ In primary SR, frequency controls the active power output, whereas, in secondary SR, the frequency and grid state is restored with additional active power.^{55,59}

The load shape changes can be achieved by executing any of the above discussed DSM strategies through a number of actions as described by Gellings.¹⁴ The characteristics of the EE programs are utility-specific and they can store all forms of RESs. The functions of the EE

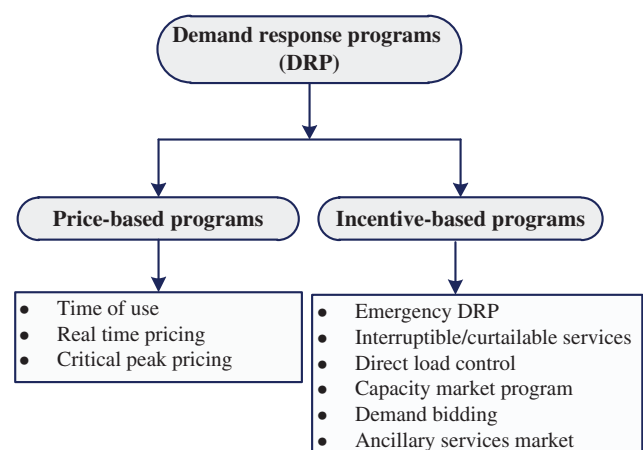


FIGURE 2 Classification of demand response programs⁶³
[Colour figure can be viewed at wileyonlinelibrary.com]

programs include the change in policies of inefficient systems, detection and replacement of misconfigured controls, motivational program for behavioral changes, and maintenance a level of consumer satisfaction.⁶² The DR programs offer financial incentives to the customers, who reduce their consumption in peak periods and different price rate at different time periods.

4 | CHALLENGES OF DSM IMPLEMENTATION

The SG system consists of several energy subsystems (Figure 3), communication, and security components.⁶⁷ The execution of DSM in SG is a challenging task and associated with a number of issues including cyber security, tariff regulation, consumer protection, adaption of modern technology and integration of RES and energy storage devices.^{26,68} The reliable operation of SG and extensive communication infrastructure depends on energy substructure, power electronics, high level information substructure, monitoring, measurement, and smart metering.⁶⁹

The cyber security and privacy are considered as key challenges, when the DSM is implemented in SG network.⁷⁰ Cyber attackers can access and misinterpret the information, which is vulnerable to the invasion of privacy. Such information includes the software of the DSM algorithm, load data, price signals, and users' personal information. In addition, attackers can easily change the load scheduling of DSM programs by introducing misinformation into the control systems; this misinformation prompts the energy supplier to refuse to respond the customers' real requests.⁷¹ Therefore, it is necessary to provide a secure and reliable operation between energy

providers and customers. To date, a number of organizations, such as the North American Electrical Reliability Corporation, Institute of Electrical and Electronics Engineers, Critical Infrastructure Protection, International Society of Automation, and US National Institute of Standards and Technology have come forward to develop rigorous solutions for maintaining the security and privacy of SG.^{72,73} Table 1 presents the highlights of DSM challenges and possible communication means and protocols to be integrated in SG network, when implementing DSM.

The lack of standard policy guidelines causes an unfair DSM implementation in SG network. The successful implementation of DSM program in SG network and analyzes of the impacts of energy savings in society require a proper supervisory action.⁸³ A proper market mechanism is needed for DSM implementation. Now-a-days DR is primarily employed for the provision of emergency contingency support and ancillary services with limited participation in the day-ahead market. This participation occurs in the form of direct market bidding as well as contracts between individual market stakeholders. The restrictive nature of these markets and contracts often requires a substantial advance notice to adjust the demand in emergency scenarios. A further barrier for DR relates to current regulatory and tariff structures, particularly for residential customers. Currently, the actual price of electricity in a customer's bill is not obvious as the final bill includes other charges such as taxes, public service obligation payments, and transmission and distribution network charges.⁸⁴

According to the researchers, fairness is another main concern for load management.^{85–89} In order to motivate the customers to shift their load, they need to be assured that they will pay a minimal amount for their electricity consumption or receive financial incentives.⁸⁷ Therefore, a suitable fairness condition should be maintained to assess the fairness of the algorithms that can help to choose an appropriate DSM program in practice. Energy system efficiency depends on the optimization of different communication protocols, SG applications, and control infrastructure.¹¹ However, due to the nature of distributed control problems and the interdependency of different domains (ie, power, communication, and control), it is entirely difficult to develop an advanced communication infrastructure. Additionally, the lack of standardization and interoperability among DSM entities inhibits the possible integration of advanced applications of smart meters, smart devices, and RESs.⁹⁰ Some DSM objectives, such as continuous interoperability, increased safety of new products and systems, robust information security, compacted set of protocols, and information exchange can be easily achieved through the standardization and interoperability of SG.⁹¹ Inaccurate time measurements and automated analysis, fast control messaging, poor visibility,

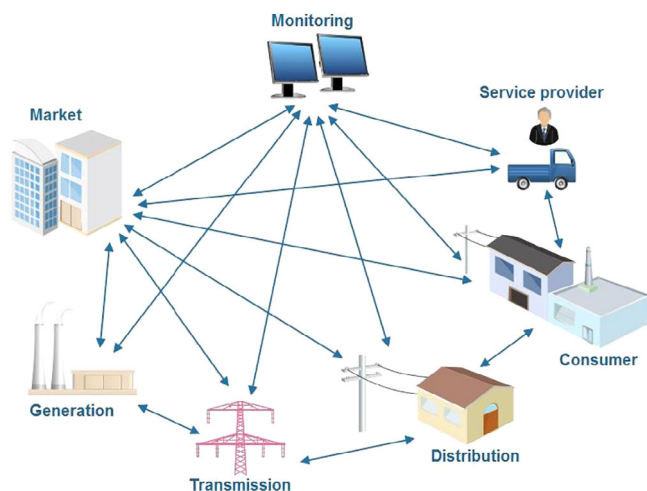


FIGURE 3 Smart grid conceptual model⁶⁶ [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 1 Challenges of DSM implementation in the SG network and possible solutions provided by different standards and protocols⁷⁴⁻⁸²

Challenges of DSM	Possible solution
Lack of reliable communication between energy sources and consumers.	Bluetooth or Ultra-Wide Band could be used for the interfaces between meter and end customer devices. IEEE 802.15.4 (ZigBee) and IEEE 802.11 (Wi-Fi) are the technical standards which could be used for smart meter interfaces in the home and local area network. Cellular wireless for example, GPRS, UMTS or 4G technologies could be used for the interfaces between meters and the central system.
Lack of interoperability among different software applications used by electric utilities	MultiSpeak, an industry-wide standards developed by the National Rural Electric Cooperative Association improves the inter-operability by defining an information model based on programming language scheme. Communication protocol focuses on web services and Simple Object Access Protocol. IEC-61850, an Inter-control Center Protocol ensures the inter-operability by specifying the definite communication networks and systems in substations.
Identification of the overall network architecture, service requirements, and device capabilities as well as ensuring the Supervisory Control and Data Acquisition and Automation Systems	Recently, the European Telecommunications Standards Institute has developed a new committee names machine-to-machine to deal with these issues. IEEE C37.1 is an IEEE Standard which deals with the system architectures and functions in a substation as well as covers the protocol selections, human machine interfaces, and implementation issues. It handles the issues related to network performance requirements, reliability, maintainability, availability, security, expandability, and changeability.
Lack of distributed resources for the implementation of DR	IEEE 1547 is a kind of standards that links the distributed energy sources with the electric power system in terms of interoperability, performance, operation, testing, and safety. DRBizNet is a DR business network which supports DR program by monitoring electricity market operations as well as creates a standard web service interface to support DR applications. This network notifies the grid operators automatically and accomplishes the specific DR program for the market. Open Automated Demand Response, a communication protocol which supports the DR program by providing a standardized information model.
Cybersecurity issues	Advanced Metering Infrastructure (AMI) Security Task Force developed a set of security requirements for AMI, which not only makes the communication transparent between utility and industry but also ensures the secure communication. These can be achieved by the transmission of security parameters, cryptographic key establishment, and management. The North American Electric Reliability Corporation provide a sets of security requirements including critical cyber asset identification, training in cybersecurity, security management controls, electronic security perimeters, incident reporting, and response planning, information protection, recovery plans for critical cyber assets and physical security. To manages the power system and secure the information exchange, IEC 62351 provides a cybersecurity requirements. They provide end to end information security by algorithms and secure manufacturing message specification. This protocol also focuses the security policies, access control, and key management of the system.

Abbreviations: DR, demand response; DSM, demand side management.

slow response times, system handling under contingency, and lack of situational awareness are also bottlenecks for effective DSM implementation.^{90,92,93} Also, rising population and demand for energy, energy storage problems, global climate change, decrease in fossil fuel sources, equipment failures, capacity limitations of electricity generation, and flexibility of problems are prime concerns according to the researchers.⁹⁴ In order to implement DSM in SG network, a number of factors such as government willingness, protocols, objective setup, funding opportunity, consumer participation, training and skill development program should be considered.

5 | INTEGRATION OF RENEWABLE ENERGY SOURCES AND STORAGE SYSTEM IN SG

RES like solar, wind, and their hybrid system has become a popular means of energy supply. However, the integration of RES with the SG has been gone through a complicated situation because of mixing a numbers of energy resources and their intermittent behavior.⁵⁰ Therefore, the DSM has been incorporated in RES integrated SG to handle the fluctuation of electricity price, the mismatch between renewable energy generation, and load demand as well as control of power transaction.^{95,96} Table 2 shows the summary of DSM implementation in RES integrated SG network. Quiggin et al modeled a residential micro-grid integrated with renewable generation technologies, energy storage, and DR programs.⁹⁹ The implementation of DR program in this model was able to reduce the peak demand fluctuations by 16% and optimize the energy balance between supply and demand. Dietrich et al analyzed the effect of DR program on the wind energy demand profile in terms of cost reduction in the system.¹⁰⁵ Incorporation of DSM program reduced the number of generation units and flattened the electricity production curve. Aghajani et al showed that utilization of DSM method reduced the effect of uncertainty, which was caused during energy generation from solar cells and wind turbine.¹⁰⁷ Çiçek and Deliç reported that the DSM method was able to achieve a steadier pattern of social welfare, which was measured in terms of customers' utility and energy generation cost.¹⁰⁸ This method maintained the balance in integration of the wind farm to the grid and dealt with the issues of energy fluctuations and economic risk. Amrollahi and Bathaee showed that DR program maintained the energy distribution in such a way that it reduced the required number of batteries, inverters, and PV cells as well as reduce the total cost.¹⁰⁹ Besides, the application of DSM improved the load factor and correlation factor by 57.9% and 36.8%, respectively. Wang et al

formulated a hybrid RES with DR program and applied to a single-family residential home.¹¹⁰ The implementation of DR scheme met the consumers' electricity demand by utilizing the available power from PV panels, wind turbine, diesel generator, and batteries. This method improved the system efficiency when compared with that of the traditional method. Behboodi et al applied DSM technique with the integration of RES to solve the multi-area electricity resource allocation problems.¹¹² This method offered an uniform electricity price by maintaining a steady-state among energy generation, transmission, and load constraints. The researchers also described an innovative approach to solve multi-area electricity resource allocation problems considering both intermittent renewables and DR. The method determined the hourly inter-area export/import set that maximizes the inter-connection surplus satisfying the transmission, generation, and load constraints. The application of DSM in SG shifts the loads from peak hours to off-peak hours, which allows storage system to store the excess power produced from the RES or in the time when grid electricity is cheap. The stored energy can be used in future when there is a shortage in the energy supply or during the peak periods or during the variability in generation from renewables. Therefore, the energy storage system (ESS) can provide a smooth real-time operation of grid as well as add economic, environmental, and technical benefits to the grid.^{113,114} The energy storage for SG can be short-term storage (ie, battery) or long-term storage (ie, compressed air, pumped hydro storage).¹¹⁵ The implementation of large capacity energy storage system in SG is a complex task, which causes high energy loss.¹¹⁶ Considering this issue, distributed energy storage (DES) system can be alternative solution excess energy storage in SG. Longe et al investigated the effect of DSM with DES in SG.¹¹⁷ A convex optimization problem was solved by energy scheduling and distributed storage algorithm in order to reduce customers' peak demand and energy cost. The proposed method was able to provide about 53% financial benefit per month to the customer. Nguyen et al formulated a centralized optimization problem to analyze the energy consumption and storage optimization problem.¹¹⁸ In this problem, each customer was connected with the energy storage device. The global optimum scheduling was achieved by non-cooperative game theoretical framework-based distributed energy consumption and storage scheduling algorithms. This method scheduled the shiftable appliance and controlled the charging and discharging of energy storage devices. The proposed method reduced the square Euclidean distance between the instantaneous electricity demand and the average demand of the power system as well as reduced the electricity payment. A dynamic game approach was proposed

TABLE 2 Impacts of DSM implementation on RES integrated SG network

Integrated system	DSM method/technique	Operation mode	Supervisory control	Outcome	Reference
Photovoltaic-battery hybrid system	DR program with Model predictive control method	Grid-connected	Centralized	Minimized the electricity bill on the customer side. Maximized the use of solar energy and battery storage.	97
Industrial microgrid with wind turbine and energy storage unit	DR scheme	Grid-connected	Centralized	Wind turbine reduced the carbon emission by 88% and DSM produced 30% more reduction. Overall electricity cost reduced by 73%	98
Residential microgrid with photovoltaic panel, wind turbine, and energy storage unit	DR scheme with linear programming method	Grid-connected	Decentralized	Energy demand was reduced by 16%. During all hours of operation, the reduction of CO ₂ emission along with the associated energy usage was 10%. During the hours of operation, the amount of renewable supply was reduced by 74%.	99
Microgrid system with photovoltaic panels, wind turbine, diesel generator, battery bank, and water supply system	DSM mechanism along with artificial neural network	Grid-connected	Decentralized	Minimized the operation cost by 3.06%.	100
Household dotted with photovoltaic systems	Load scheduling method based on online event-triggered energy management algorithm	Grid-connected	Centralized	Reduced the electricity bill as well as ensured the user comfort level.	101
SG network with renewable distributed generators	DR scheme with parallel autonomous optimization	Grid-connected	Centralized	Reduced electricity generation cost and electricity bill.	102
Microgrid system with micro turbines, wind turbine, fuel cells, photovoltaic panels, storage devices and a group of radial load feeders	DR scheme	Grid-connected	Centralized	Peak load was shaved from the grid tie-line. Achieved optimal scheduling of batteries and diesel generators.	103
Microgrid with renewable generators and energy storage	DR scheme	Isolated	Centralized	Achieved an optimal power generation and peak load dispatch.	104
SG network with high wind penetration	DR scheme	Isolated	Centralized	Achieved 30% cost savings. More than 56% of demand was shifted.	105
SG network with the energy storage device	Load scheduling with game theory algorithm	Grid-connected	Centralized	Reduced peak load as well as energy payment for the consumers.	106
	DR scheme	Grid-connected	Decentralized		107

TABLE 2 (Continued)

Integrated system	DSM method/technique	Operation mode	Supervisory control	Outcome	Reference
Microgrid network with wind turbine and solar cell				Reduced the operational cost and carbon emission.	
SG network with wind farm	DR scheme	Grid-connected	Centralized	Achieved optimal scheduling of energy production and consumption for 24 hour.	108
Microgrid system with photovoltaic system, wind turbine, and battery	DR method with mixed-integer linear programming	Isolated	Decentralized	Operational cost and peak load were reduced by 17.2 and 36.8% respectively.	109
Energy system with photovoltaic panels, wind turbine, diesel generators, and batteries	DR scheme	Grid-connected	Centralized	Reduced operational cost and environmental cost.	110
SG network with photovoltaic system	DR scheme	Grid-connected	Decentralized	Load factor is analyzed and increased during a year.	111

Abbreviation: DR, demand response.

by Pilz and Al-Fagih for scheduling the energy storage for the residential household.¹¹⁹ An extensive analysis was carried out based on forecasting error. The proposed method proved as beneficial for both utility and customer in terms of gain. Aktas et al proposed a hybrid ESS consisting of battery bank and ultra-capacitor unit to ensure high power density ability during sudden fluctuation of energy demand.¹²⁰ Shen et al also suggested a hybrid ESS for a DC micro-grid system for electric vehicle charging station.¹²¹ The hybrid ESS comprised of a flywheel and a lithium iron phosphate battery. However, real-time extensive research is further required on the integration of ESS in SG and optimization of their key performance parameters.

6 | PROGRESS OF DSM OPTIMIZATION MODELS AND APPLICATIONS OF ALGORITHMS

The existing literature indicates that various types of algorithms including single and hybrid ones have been developed and implemented to solve the optimization problems of the DSM of SG. The particle swarm optimization (PSO), GA, game theory algorithm (GTA), ACO, linear programming (LP), non-LP (NLP), and dynamic programming (DP) are the most widely studied algorithms in the field of DSM. Recently, hybrid algorithms have gained remarkable attention as promising methods. Table 3 shows the characteristics and user-defined parameters of various algorithms used in DSM. This section provides an overview of the research and advancement of different algorithm-based optimization methods for DSM.

6.1 | Optimization models with GA

The GA is a metaheuristic and an adaptive search method inspired by the process of natural selection, which is a biological mechanism. In 1960-1970, the GA was established by John Holland, which was used to solve search and optimization problems by natural evolutions, such as inheritance, mutation, selection, and crossover.^{122,142-146} Figure 4 depicts the flowchart of a typical GA.

The GA has been applied to many optimization problems to achieve the desired objectives of DSM. Table 4 summarizes the algorithms used for solving the DSM optimization problems in SG. The reviews related to the application of the GA in DSM are as follows. Logenthiran et al proposed a day-ahead load shifting technique for the DSM of SG.⁴⁰ They used a heuristic-based evolutionary algorithm to solve a minimization problem and shift the load from peak hours to off-peak hours. The simulations were carried out in residential, commercial, and industrial areas

TABLE 3 Characteristics of various algorithms used in demand side management

Algorithm type	Algorithm name	Mechanism	User-defined parameters	Characteristics	Reference
Metaheuristic and evolutionary algorithm	GA	Inspired by the mechanism of natural selection.	The size of the population of solutions, the number of parents, the probability of crossover, the probability of mutation and the termination criterion.	Genes of chromosome represent the decision variable. This variable contains binary, continuous or discrete values. Genetic operators are responsible for the creation of new solutions. Individual chromosome provides a possible solution and parents provide an old solution while a new solution is provided by offspring. Elite provides the best solution. Population diversity and selective pressure affect the search method. Correction of convergence depends on the selection of a good termination criterion and optimum selective pressure.	122-124
	Particle swarm optimization	Inspired by the social behavior of birds flocks.	Size of the population of solutions, the value of the initial inertia weight, the final value of the inertia weight, and the termination criterion.	The decision variable is represented by the particle position in each dimension. The solution of the optimization problem is found by the position of the particle where the position is updated to find a new solution. Fitness function is measured by the distance between particle and food. A number of iterations, selection of good termination criteria, the improvement of the objective function, and the run time of the algorithm determine the confection of convergence.	123,125-128
	ACO	Inspired by the collective and searching behavior of ant species.	The size of the population of solutions, the evaporation rate, the control parameters of pheromone, heuristic information, and the termination criterion.	Decision variables are represented by the path of an ant. In the case of an optimization problem, a possible solution is determined by the tour of an ant from nest to food. Process of generating new solutions is accelerated by the information-based stochastic mechanism.	123,129-132

TABLE 3 (Continued)

Algorithm type	Algorithm name	Mechanism	User-defined parameters	Characteristics	Reference
Classical method	LP	Mathematical programming method where the objective function is linear.	The collection of coefficients with respect to decision variable, constraint, upper bound is the parameter of the LP method.	ACO allocates desirability to the decision space according to the fitness value of a solution.	133-136
				Correction of convergence depends on the number of iterations, selection of good termination criteria, the incremental improvement of the objective function, and the run time of the algorithm.	
				Objective functions are correspondents to a restricted set of constraint. It has a feasible solution and region. The optimal solution can be found. Multiplicity in solutions.	
Classical method	Non-LP	Mathematical programming with respect to the nonlinear objective function.	Parameters are defined based on the problems	Converting a complex problem into an easy problem. Solving the sequence of sub-problems. Solving of sub-problems are involved with the unconstrained minimization function. The optimal solution can be found.	137-140
				Representing the multistage decision process. For each stage, a policy decision is requested. Solving multivariable optimization problem. In order to determine the optimal solution for the problem, the solution method is categorized. Recursive relation is used to optimize the solution procedure.	
Classical method	DP	Multistage nature of optimization method.	There is no specific parameter. Each problem has its own parameter.		140,141

Abbreviations: ACO, ant colony optimization; DSM, demand side management; LP, linear programming.

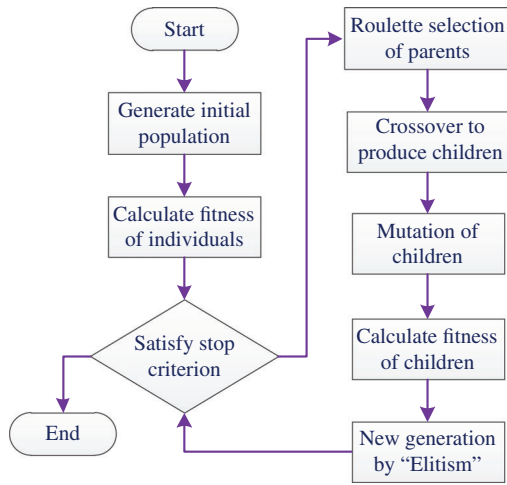


FIGURE 4 Flowchart of the genetic algorithm¹⁴⁷ [Colour figure can be viewed at wileyonlinelibrary.com]

with a variety of loads. In this work, the authors considered the central controller of the SG to control the DSM technique. However, if the controller becomes disabled for any reason, this whole process will stop functioning. Compared with other areas, residential areas showed the highest number of devices available for control. However, the amount of reduction in operating cost was not as expected because no incentive scheme or financial reward plan was considered to motivate the customers to shift their load. In another study, a GA-based DSM method was proposed to solve the objective function.¹⁵⁸ In this case, the method considered only industrial load. The authors used a load shifting technique to reduce customers' inconvenience, energy generation cost, and total electricity cost. Similarly, Bharathi et al applied a heuristic-based GA to model the DSM.¹⁵⁷ The proposed DSM model reshaped the load patterns and reduced energy usage in industrial, commercial, and residential areas by using a suitable load shifting technique. Arabali et al introduced a GA-based SG strategy for shifting residential cooling loads to match renewable energy production.²⁴ The authors recommended to use the developed approach in heating, ventilation, and air conditioning loads. Yao et al developed a DSM model based on a modified GA, named iterative deepening GA, to optimize the scheduling of DLC approaches and minimize the revenue loss of electricity generation companies.⁴⁸ They considered only the air conditioning load, which has a low impact. Therefore, various types of loads need to be considered for further justifying the iterative deepening GA.

6.2 | Optimization models with PSO

The PSO algorithm is an evolutionary computational method that was first introduced by Dr. Eberhart and

Dr. Kennedy in 1995.^{159,160} PSO is a population-based optimization method inspired by the social behavior of bird flocking or fish schooling. The PSO algorithm first creates the initial particles and assigns them initial velocities. Then, it calculates the objective function at each particle location, determines the best (lowest) function value, and identifies the best locations. PSO has two updated functions, namely, velocity updated function and position updated function. The velocity of a particle can be mathematically modeled as follows.¹⁶¹

$$V_i^{t+1} = w \cdot V_i^t + C_1 \cdot \text{rand}() \cdot (x_{pbest-i}^t - x_i^t) + C_2 \cdot \text{rand}() \cdot (x_{gbest-i}^t - x_i^t),$$

where, $x_i^{t+1} = x_i^t + V_i^{t+1}$.

In this case, V_i^{t+1} is the velocity of agent i at iteration $t+1$, w is the weighting function, C_1 and C_2 are the weighting factors, rand is the uniformly distributed random number between 0 and 1, x_i^t is the current position of agent i at iteration t , $x_{pbest-i}^t$ is a coordinate of the best known result in the swarm, and $x_{gbest-i}^t$ is a coordinate of the best known result in the swarm. Figure 5 illustrates the typical flowchart of the PSO algorithm.

The PSO algorithm-based techniques have been widely implemented by researchers to solve energy management problems. The PSO algorithm is considered as significantly effective in solving various optimization problems.¹⁶³ The PSO-based techniques are reviewed in the recent study. Logenthiran et al modeled a day-ahead load scheduling technique that incorporates the PSO algorithm based on the customers' inputs and forecasted hourly electricity rates.¹⁵⁰ In this study, the authors considered the shiftable and non-shiftable loads controlled by a central controller of the SG. The simulation was carried out in residential, commercial, and industrial areas, and the results revealed a reduction in PAR and an increase in electricity cost savings. However, no incentive scheme or reward plan was applied to compensate the customers for giving up their comfort zone and shifting their load from peak hours to off-peak hours. Nayak also developed a PSO-based DSM strategy that considers a load shifting technique mainly for residential loads.¹⁶⁴ The methodology mainly comprises population-based heuristic optimization techniques which are used to solve the scheduling problems and provide global optimum solutions.

The multi-objective PSO algorithm was studied by Aghajani et al for reducing the operating cost and emission with the integration of renewable sources in the micro-grid.¹⁰⁷ They recommended a stochastic programming that focused on probability density functions and was integrated with the DR model to optimize the

TABLE 4 Summary of results of the algorithms used in DSM

Algorithm	Methodology	Reduction of PAR (%)	Minimization of energy cost (%)	Software	Remarks	Reference
GA	Heuristic-based load shifting technique	14.2-18.3	Customers' savings: 5-10 Company's savings:15-20	Real-time simulation	The algorithm converged well and globally.	40
Multi-objective PSO method based on the fuzzy technique	Probabilistic model-based on incentive payment demand response programs	×	×	MATLAB	Pollution emission factor was considered.	107
GTA	Energy consumption scheduling approach	38.1	37.8	MATLAB	The algorithm converged locally.	148
GTA	Distributed energy storage planning	31.5	22.43	MATLAB	The algorithm converged globally with minimum information exchange.	149
PSO	Load shifting technique	43	18	MATLAB	Algorithm converged globally and required less time interval.	150
Non-stationary DSM algorithm based on a repeated game framework	Incentive compatible	10	50 (including discomfort and billing cost)	Real-time simulation	A simulation was carried out in the homogeneous and heterogeneous situations. The algorithm converged globally.	46
Vickrey-Clarke-Groov (VCG)	VCG pricing method	19.3	37.8	MATLAB	The algorithm converged globally with 50 users. Shifting and executing the algorithm required a certain amount of time.	151
Distributed algorithm	DSM scheme based on time-varying pricing	64.76	×	MATLAB optimization solver Mosek	The residential load was considered, and a simulation was carried out on 35 appliances.	152
GTA	Autonomous and distributed DSM scheme	17	19.6	MATLAB	The algorithm converged locally and required minimal execution time.	153
Non-linear mixed-integer linear programming	Dispatching model with DR	×	Reduced	MATLAB	The optimal dispatching model was unable to handle large loads.	154
ant colony optimization	Congestion management method	×	Reduced	MATLAB	The integration of RESs was not considered.	155

(Continues)

TABLE 4 (Continued)

Algorithm	Methodology	Reduction of PAR (%)	Minimization of energy cost (%)	Software	Remarks	Reference
Distributed algorithm	Autonomous energy scheduling scheme	24	21	MATLAB	The algorithm converged locally.	156
Sequential Gauss-Siedel algorithm	Parallel autonomous optimization scheme with DR framework	19.71	5.53	Real-time simulation	The algorithm converged locally.	60
GTA	Energy consumption and storage optimization method	40	19	Real-time simulation	The algorithm converged in parallel. The algorithm was able to handle a large number of users and simultaneously update their strategies.	106
GA	Load shifting technique	23.84	×	MATLAB	The algorithm performed well.	157

Abbreviations: DR, demand response; DSM, demand side management; GTA, game theory algorithm; PSO, particle swarm optimization.

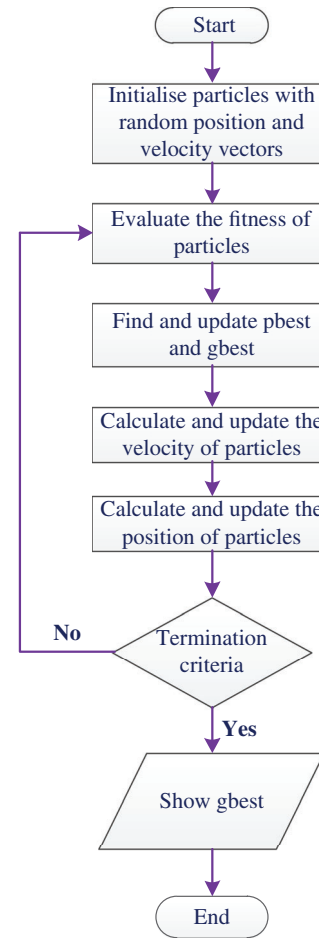


FIGURE 5 Flowchart of particle swarm optimization algorithm¹⁶² [Colour figure can be viewed at wileyonlinelibrary.com]

performance of smart micro-grids. However, the uncertain behavior of predicted power generation from wind turbines and solar cells remarkably affect the operation cost. Therefore, some existing studies focused on the implementation of BPSO in DSM, which is a modified form of the PSO algorithm. Kennedy and Eberhard first employed the BPSO algorithm to schedule interruptible loads and solved a multi-objective optimization problem.¹⁶⁵ Pedrasa et al also suggested the implementation of BPSO to optimize the DSM problems.¹⁶⁶ A realistic scheduling mechanism based on the BPSO for SG was suggested by Mahmood et al.¹⁶⁷ They found the BPSO algorithm to be an effective algorithm for reducing electricity costs. In this case, the appliances were categorized according to the respective constraints and effective time of usage for increasing the appliances' utility. Zhou et al proposed a real-time optimal appliance usage approach to maintain energy usage based on the BPSO algorithm.¹⁶⁸ The method smoothed the peak shaving, valley filling, and demand curve as well as reduced energy usage with

the assistance of customers and energy suppliers. Zhou and Xu also applied the BPSO algorithm to solve the cost function of electric vehicle users, SG, and power sources.¹⁶⁹ The simulations for load shifting, energy-saving, and energy supply efficiency were carried out in the MATLAB platform.

6.3 | Optimization models with ACO

The ACO algorithm was proposed by Macro Dorigo in 1992.^{170,171} The algorithm aimed to search for an optimal path in a graph. Principally, the algorithm is used to solve discrete combinatorial optimization problems. The algorithm presents some unique properties, such as self-healing, self-protection, and self-organization.¹⁷² The flowchart of the ACO algorithm is given in Figure 6.

Extensive research based on the implementation of ACO algorithm has been performed to handle energy management optimization problems. Dethlefs et al used a distributed ACO-based self-optimization method for producing a day-ahead schedule.¹⁷² In their work, only shiftable loads were considered to reduce the external purchase of energy. The algorithm optimized the load of distributed consumers and the power generated from a wind power plant. In this case, almost 10% of the power rating was used to control the residential shiftable appliances, and the algorithm mainly adjusted the mean generation and demand. An efficient DSM model based on the ACO

was presented by Rahim et al to control residential energy.¹⁷⁴ The model reduced the peak load, PAR, and electricity costs considering the customers' satisfaction levels. They also used a TOU tariff model with an inclined block rate to avoid the peak load and complexity in the estimation of electricity bills. Hazra et al explored an efficient method for handling the load congestion problems in SG in an economical way.¹⁷⁵ The problem was solved by using the DR, ACO, and fuzzy techniques. Their findings suggested that the method was able to reduce the electricity cost and fulfill customers' satisfaction through the scheduling of different generation resources. In another study, the authors analyzed the load congestion management problems to control model cost.¹⁵⁵ The problem of real-time congestion management was developed as an NLP problem. In this study, the ACO algorithm provided a feasible solution for the problem and minimized the electricity cost. Okonta et al proposed an ACO-based load scheduling algorithm for a smart home.¹⁷³ The researchers mainly focused on the total electricity bill, TOU, and the overall increase in quality of life with the incorporation of the optimal utilization of integrated RESs. An automated load manager based on the ACO algorithm and an interactive web interface were used for DLC and energy management to allow users to access their home appliances from remote areas via the internet.

6.4 | Optimization models with GTA

GTA is a strategic decision-making tool for situations involving several players aspiring for different purposes. Generally, game theory refers to the study of mathematical models of conflict and cooperation between decision-makers.^{176,177} It is a formal analytical method of strategic decision making, which is embedded with a set of mathematical models for the study of complex interactions among independent rational players.¹⁷⁸⁻¹⁸⁰ Typically, a game can be defined as a situation consisting of the following three components.¹⁸¹

- A set of players N
- For every player, a set of possible strategies $(s_i)_{i \in N}$
- For every player, a utility function $(u_i)_{i \in N}$

Figure 7 shows the flowchart of the GTA applied to the modeling of DR for residential consumers in an SG environment.

Recently, the GTA algorithms have gained remarkable popularity in solving the DSM optimization problems because of their capability of solving distributed system problems. In addition, designing an algorithm using game theory is relatively easy.¹⁸³ Hung Khanh

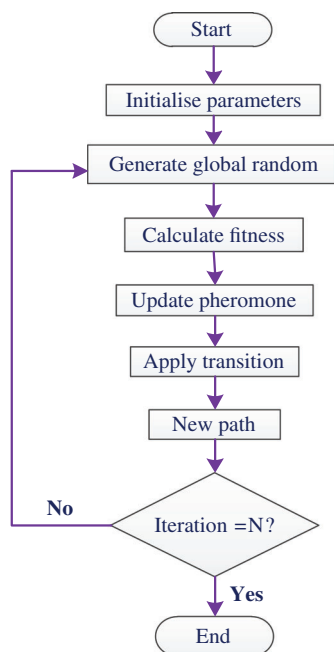


FIGURE 6 Flowchart of the ant colony optimization algorithm¹⁷³ [Colour figure can be viewed at wileyonlinelibrary.com]

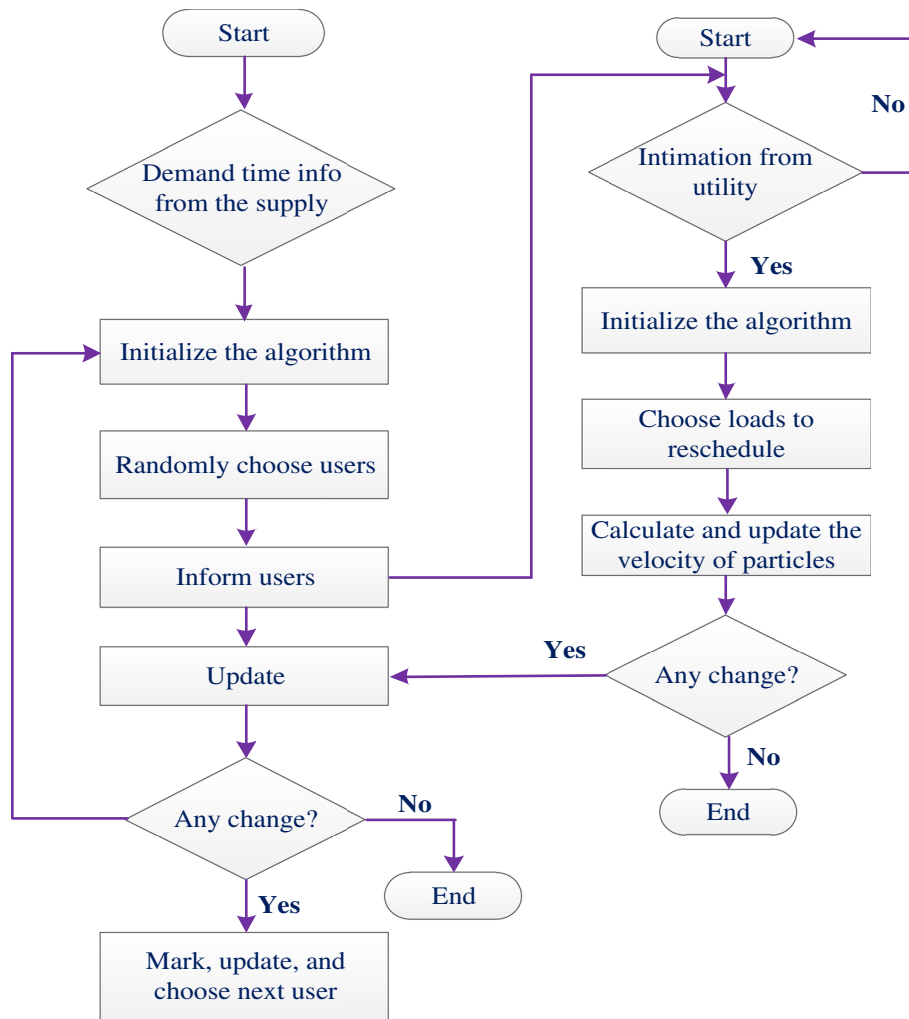


FIGURE 7 Flowchart of game theory algorithm between utility and user¹⁸² [Colour figure can be viewed at wileyonlinelibrary.com]

Nguyen et al applied a non-cooperative game theory model to formulate the DSM problems.³⁴ The model reduced the peak demand, PAR, and total energy cost. In this case, the GTA was not able to converge for the optimal solution in the centralized design. Nevertheless, the algorithm minimized the PAR, close to the optimal solution of the centralized design. Here, the impacts of large numbers of users with battery on system performance were not investigated. The large numbers of users integrated with battery can influence the aggregate cost and PAR reduction. Song et al⁴⁶ developed an optimum non-stationary DSM model based on a repeated game model, which was mainly an incentive-compatible model. This model was designed to allow consumers to choose their daily consumption patterns independently without affecting their habits, choices, and wants. Under this strategy, an active set of consumers was selected based on the historical energy consumption patterns and the billing cost compensation was considered to motivate the consumers. Wang et al studied a DSM model integrated with cognitive radio technology based on the proficient and trustworthy communication infrastructure in SG.¹⁴⁹ This

model suggested a cost function focusing on customers' preference with a balanced payment model consisting of billing, electricity generation, and discomfort costs. This DSM system was designed to allow users to select an appropriate size of storage units for balancing the costs. In this study, the GTA was applied to optimize the distributed planning storage method. The results indicated the reduction of PAR, total energy cost, customers' daily payment, and energy consumption. This work also proved that the application of cognitive radio technology can effectively reduce energy consumption in the SG communication networks. However, customers' privacy was not guaranteed in this case. Deng et al analyzed the residential energy consumption scheduling problem and formulated a couple constrained game with respect to interactions among customers.¹⁸⁴ They applied a real-time pricing approach that shifted the peak demand to the off-peak hours to balance the energy demand. A GTA-based autonomous and distributed DSM scheme was proposed by Mohsenian-Rad et al,¹⁵³ who focused on the scheduling of energy consumption with the consideration of residential loads. The proposed technique was

based on incentives and thus reduced the peak load, total energy costs, and customers' daily electricity charge. In this investigation, the authors employed a new energy cost function; however, the estimation of energy consumption and simulation was time consuming. In another research, Nguyen et al proposed a smart power system with an energy storage device based on the GTA.¹⁰⁶ The objective of the study was to diminish the square Euclidean distance between the instantaneous and regular electricity demands of the energy system. The power consumption was scheduled using the energy cost-sharing model, and the loads were synchronized by a principal controller.

6.5 | Optimization models with LP

LP or linear optimization method is used to represent mathematical models by linear relationship to achieve the best results, such as maximum profit or minimum cost.^{133,185,186} It optimizes a linear objective function with respect to linear equality and linear inequality constraints. The general formula of a related optimization problem can be expressed as follows:

$$\text{Minimize } f^T x$$

$$\text{Subjected to } A \cdot x \leq b \text{ and } x \geq 0.$$

Figure 8 illustrates the typical flowchart of the simplex method for LP.

Many researchers have applied the LP method to solve DSM optimization problems and their results are summarized in the current work. Sheblt studied a load management scheduling program using the LP method.¹⁸⁸ A DLC scheme was used to schedule appliances and to increase the profit of energy suppliers according to cost or market price function. Kurucz et al also applied the LP model for scheduling the loads under control periods to minimize the system peak load.¹⁸⁹ The LP model considered the residential, commercial, and industrial loads to bring a specific number of customers under the model and determine long-term and short-term control scheduling strategies. An integer LP-based load scheduling mechanism was proposed by Zhu et al for the DSM of SG.¹⁹⁰ With the aim of reducing peak hour loads, they used the proposed mechanism to schedule home appliances along with the optimal power and operation time according to customers' preferences. Martins et al proposed a multi-objective LP model to increase the power generation capacity.¹⁹¹ The total extension cost, the environmental impact associated with energy output, and the environmental effects associated with the installed power capacity were considered in the

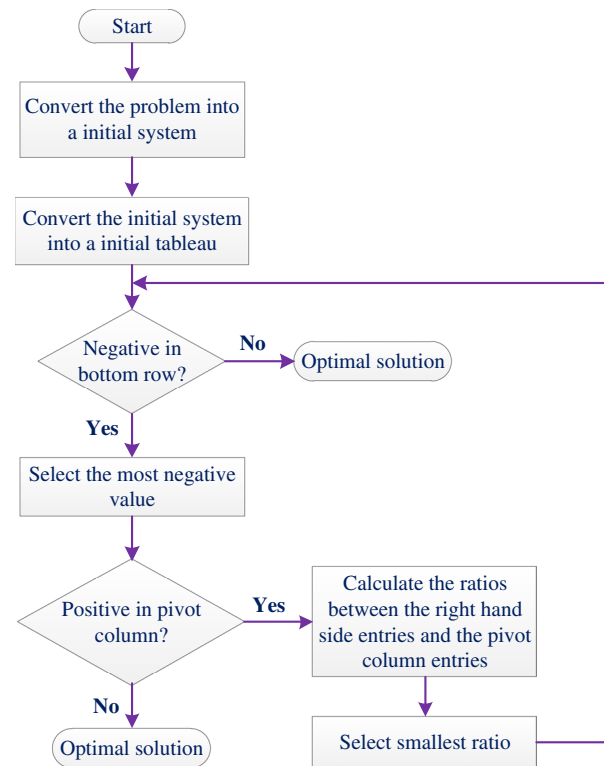


FIGURE 8 Flowchart of the simplex method for linear programming¹⁸⁷ [Colour figure can be viewed at wileyonlinelibrary.com]

investigation. In order to solve the objective function, they also considered five constraints, such as the reliability of the supply system, the availability of generation units, and the capacity of the generation group equivalent to the DSM.

6.6 | Optimization models with NLP

NLP is used to solve mathematical optimization problems in which some of the constraints and objective functions are nonlinear.^{192,193} In the NLP method, the minimized or maximized objective function can be expressed in the following form¹⁹⁴:

$$\text{Minimize or maximize } f(x),$$

$$\text{Subjected to } g_i(x) \leq 0 \text{ for each } i \in \{1, \dots, p\},$$

$$h_j(x) = 0 \text{ for each } j \in \{1, \dots, q\},$$

$$x \in X,$$

where, x is a subset of R^n ; and f , g_i , h_j are real valued functions with $i = \{1 \dots p\}$, $j = \{1 \dots q\}$, in which one of the

real valued functions are nonlinear. m , p , and q are positive integers. Figure 9 shows a simple NLP flowchart.

Many researchers have used the NLP method in DSM because it improves cost functions and generates satisfactory results.¹⁹⁶ Shaaban et al used a mixed-integer NLP (MINLP) method for scheduling batteries and shiftable and adjustable loads.¹⁹⁷ The implementation of this energy management technique was able to reduce the operating cost under an SG network. Wang et al also used the MINLP method to optimize the optimal dispatching model of a smart HEMS with distributed energy resources and intelligent domestic appliances.¹⁵⁴ This method reduced electricity costs and aggregated power consumption. However, the MINLP was not able to handle many appliances because of the unpredictable, impulsive, non-linear, and complex energy consumption patterns of consumers. Considering DR, Helal et al proposed a MINLP-based energy management model to optimally schedule the different generation technologies of AC/DC hybrid micro-grids in islands.¹⁹⁸ They also suggested that the system depends on a micro-grid controller, which ensures the proper usage of energy with minimum operating cost by controlling user appliances and water desalination units. An optimum schedule with minimum cost was achieved by formulating the scheduling problems as MINLP problems.

6.7 | Optimization models with DP

The DP method is a mathematical optimization method that divides the problem into smaller sub-problems. This method can solve sequential or multi-stage problems.¹⁹³ In the DP method, the mathematical expression of a multi-stage decision process can be written as follows.^{199,200}

$$c_n = f_n(b_n, s_n),$$

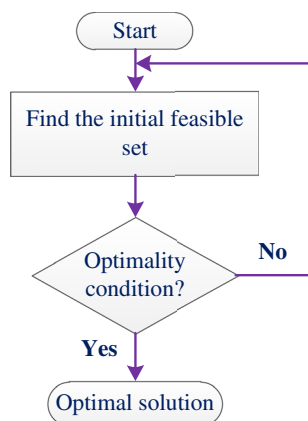


FIGURE 9 Flowchart of simple non-LP¹⁹⁵ [Colour figure can be viewed at wileyonlinelibrary.com]

where, c_n is the cost for a particular stage n , b_n is a permissible decision, and s_n is the state of the process with n stages.

Thus far, this model has been used in several studies to solve the DSM optimization problems of SG. Chu et al proposed a DP-based optimization algorithm to determine the scheduling schemes of DLC.²⁰¹ The authors considered an air conditioner in a commercial building for load scheduling. The method reduced the peak load according to customers' discomfort level. In addition, this method was found to be effective in scheduling other appliances by determining a target load level and controlling load usage to reduce the peak load and electricity generation costs.²⁰² Primarily, an analytic DP model was used to schedule some of the appliances of residential loads. In this study, the performance of the algorithms and the control periods of appliances were investigated in five cases; however, the satisfaction levels of customers were not considered. Hsu et al introduced a DP-based optimization method for reducing the system's energy generation cost for the DLC dispatch.²⁰³ Therefore, the DLC strategies were integrated with a unit commitment problem, and a DP method was developed to solve dispatch DLC and the unit commitment problem. To reduce the electricity demand in the SG environment, Reka and Ramesh proposed a DP model with a cloud computing framework,²⁰⁴ which created a small energy hub with customers and displayed the customers' participation in DSM programs.

6.8 | Optimization models with emerging techniques and methods

In the above subsections, most commonly used algorithms in the DSM of SG are reviewed. In contrast to these algorithms, some emerging approaches, including hybrid methods, have been proposed for solving the DSM optimization problems. However, studies related to the implementation of these approaches are limited in the literature. Thus far, the bacterial foraging optimization (BFO) algorithm, gray wolf optimization (GWO) algorithm, wind-driven optimization (WDO) algorithm, enhanced differential evolution (EDE), and harmony search (HS) algorithm have been investigated for solving DSM optimization problems in SG.

Esther et al used a BFO-based load shifting technique to solve a minimization problem.²⁰⁵ The proposed BFO algorithm was able to handle the different types of loads in residential, commercial, and industrial areas. The authors reduced the peak load and energy cost by 7% to 10%. The BFO algorithm showed better performance than the heuristic-based evolutionary algorithm. Moreover, the BFO

algorithm was able to handle a large number of controllable devices to achieve the desired objective function. Barolli et al used GWO and BFO to evaluate the performance of HEMS.²⁰⁶ In this work, home appliances were categorized into two classes based on their power consumption patterns. The BFO and GWO reduced the energy cost by 45% and 55%, respectively. Furthermore, Zafar et al measured the performance of a HEMS using three meta-heuristic optimization techniques, namely, the HS algorithm, BFO algorithm, and EDE algorithm.²⁰⁷ The performance of the algorithms was measured in terms of PAR reduction and energy cost minimization. The HAS algorithm achieved higher energy cost and PAR reduction than the BFO algorithm. The existing optimization methods were non-flexible in nature and were found to be unsuitable in terms of handling the complexity of cost minimization and user comfort maximization problems.^{20,24,168,208,209} These nonlinear problems create a high computational complexity. Similar to the GA, the BPSO, WDO, and BFO algorithms can solve multi-objective optimization problems. Moreover, the WDO and BFO algorithms were found to be similar to the PSO algorithm as population-based search methods.²¹⁰

Some modified algorithm and hybrid approaches (a combination of two or more algorithms) have been considered as promising techniques to overcome the limitations of single algorithms. Compared to a single algorithm, some of the hybrids perform better in terms of PAR reduction, electricity bill reduction, users' discomfort, and cost savings. For instance, the hybrid GA and PSO were found to be highly efficient when handling different nonlinear optimization.²¹¹ Javaid et al developed a hybrid genetic wind-driven (HGWD) algorithm to design a DSM controller for a residential area in an SG.²¹² The result showed that the HGWD algorithm performed the best. For example, the HGWD algorithm reduced the electricity consumption cost by 33% and 10% when compared with the WDO algorithm and GA, respectively. The HGWD minimized the user comfort to 40% and achieved an optimal result with 17% reduction of PAR and 30% minimization of electricity bill. Ahmad et al proposed a hybrid algorithm consisting of GA and PSO (HGPSO) and showed that HGPSO performed better than the GA, BPSO, BFO, WDO algorithms.²¹³ The percentage of PAR reduction for the GA, BPSO, BFO, and WDO algorithms were 14.09%, 3.30%, 22.10% and 33.54%, respectively and the percentage of electricity bill reduction were 9.80%, 19.50%, 15.40% and 15.80%, respectively. On the other hand, the HGPSO reduced the electric bill and PAR by 25.12% and 24.88%, respectively. Another study proposed an improved PSO algorithm (IPSO) and compared with the GA.²¹⁴ The GA reduced the peak load by 25.78% while almost 30.26% of peak load was reduced with the IPSO. Manzoor et al suggested a hybrid method called the

teacher learning genetic optimization (TLGO) algorithm for residential load scheduling with day-ahead pricing scheme and compared with the teacher learning-based optimization (TLBO) and GA.²¹⁵ The cost savings with the GA, TLBO, and TLGO were 31%, 31.5%, and 33%, respectively. The TLGO had the lowest user discomfort compare to GA and TLBO. The value of user discomfort with the GA, TLBO, and TLGO were 2.37, 2.14, and 1.83, respectively. In addition, Zhang et al proposed a hybrid algorithm based on the EDE and HS algorithms to design a day-ahead scheduling model for microgrid systems with the integration RESs.²¹⁶ The hybrid approach minimized the start-up cost and generation cost of the RESs, as well as improved the tuning parameters. Javaid et al developed a hybrid algorithm named bat-crow search algorithm (BCSA) with the combination of a meta-heuristic bat algorithm (BA) and crow search algorithm (CSA).²¹⁷ They used critical peak pricing (CPP) scheme for HEMS and compared the result of BCSA with BA and CSA in terms of reduction of electricity cost. Optimization result showed that the BCSA algorithm reduced electricity cost by 31.19% while the BA and CSA reduced by 28.32% and 26.70%, respectively. It can be concluded from the above discussion that the hybrid algorithms perform better over single algorithms due to the higher search efficiency and flexibility of hybrid algorithms compared to single algorithms.

7 | COMPARATIVE DISCUSSION OF OPTIMIZATION APPROACHES

The techniques and strategies used in DSM have already been found with immense potential to improve power systems by handling and controlling loads. However, several issues are associated with their implementation in DSM, and these issues need to be resolved to enhance their contribution to the improvement of power systems. The previous discussion explains that most of the existing researches are not based on the real-time applications. With the application of real-time DSM strategies and techniques such as DR, load shifting, and load curtailment can be established for the real-time operation of SG.⁴⁰ The real-time execution of DSM is much possible, particularly in an artificial intelligent platform provided by a multi-agent system.²¹⁸ Moreover, the application and utilization of SG requires some computational efficiency, which can be obtained by using a cloud computing framework. The purpose of using a cloud computing framework in the application of DSM in SG is to maintain the scheduling mechanism for the DR program in real-time, ensure security and privacy for the SG, and analyze the data properly.¹⁵⁷

The algorithms such as GA, PSO, ACO, LP, NLP, and DP used for solving the DSM optimization problems are different in terms of efficiency, reliability, simplicity, and convergence speed as shown in Table 5. Among the algorithms, the GA, PSO, and ACO are meta-heuristic and evolutionary techniques that explore the different areas of search space to obtain an optimum solution. In addition, the GA, PSO, and ACO exhibit high system independence and efficiency. The incorporation of a naive approach can improve the efficiency of the GA in handling fixed and large numbers of iterations, but the algorithm can provide misleading results.²³⁹ The implementation of PSO algorithms is difficult because of their complexity, poor dynamic response, and long tracking process.²⁴⁰ The ACO can model an optimization problem to determine a minimum cost path.¹²³ In this case, the number of decision variables in the path determines the solution of the optimization problem. The pheromone evaporation and weighting parameters are the standard parameters of ACO. In contrast, the GA considers crossover probability, mutation probability, and population size.²⁴¹

Swarm size, neighborhood size, number of iterations, and acceleration coefficients are the basic parameters of PSO. Before executing the algorithm, the parameter must be set. Therefore, the execution of many time-consuming experiments is needed to define a good set of parameters. The GA, PSO, and ACO have a fast convergence speed. However, the GA cannot easily converge in the presence of noise. The ACO shows a fast convergence speed relative to the GA as reported in the previous literature.²⁴²

A framework based on game theory is used to interact among multiple primary and secondary users and analyze network dynamics with different system parameters. Similar to the GA, the GTA includes several players, strategies or actions, orders, and payoff functions.²⁴³ The game algorithm is more suitable compared to the optimization algorithms for studying frequency-dependent selection. In addition, the GTA is an effective method for implementing DR programs by facilitating the interaction between power utility companies and users and simplifying intelligent decision making in DR frameworks.¹⁷⁹ The DR model based on game theory converges to Nash or other equilibriums to obtain an optimal solution for supply and demand sides. Game-theory based designs in control systems face the risk of duplication because of errors in the decision-making process. Moreover, the increase of player number in the GTA increases the complexity level.

The LP method is applicable to problems with a linear objective function and constraint. The LP method provides a high solution speed and exhibits high reliability with the possibility of achieving optimal solutions.¹³³ The linearized problem made from a nonlinear one may sometimes cause inaccuracies which leads to inaccurate

models. Therefore, the LP method needs to be improved to maintain grids' stochasticity, predictivity, adaptivity, and randomness. This improvement is based on the limitations of the conventional LP method for static problems that can create problems in SG implementation. On the contrary, the NLP method finds candidate solutions for the local or global optimum. The constraint satisfaction of NLP includes the optimization and evaluation of the optimum value of an objective function. Similar to the LP method, the NLP method cannot handle variability and predictivity. The NLP method also has large computational burden and limitation to static variables in the objectives and constraints. The DP method provides a solution for sequential multi-stage and decision problems via the candidate optimization technique.²⁴⁴ Recursive optimization helps to achieve the overall optimum. The DP method handles objective and constrained optimization problems in the presence of time variability and noise. However, this method suffers from dimensionality, which is not suitable for large and complex power systems.¹⁹⁵

The hybrid and modified algorithm were found to be performed better and provided better result compared to the single algorithms. The hybrid method TLGO reduced the energy usage cost and functioned as a method with low computational complexity relative to the LP.²¹⁵ In addition, the hybrid differential evaluation PSO algorithm was found to overcome the limitations of the EDE and PSO algorithms, such as slow convergence and lack of randomness.²⁴⁵ In the case of DSM, the HGPO uniformly distributes the load without creating any other peak and the appliances have minimum delay compared to other heuristic algorithms.²¹³ The IPSO has good convergence speed rather than the GA and quickly adopts the new strategies in terms of initialization of user-defined parameter.²¹⁴ The EA algorithm has to face premature convergence and slow convergence issues to solve a global and nonconvex optimization problem. Kim et al proposed a hybrid method combining Euclidian distance (EU) based GA and PSO (EU-GA-PSO) to find the global and local solution.²⁴⁶ This method had a faster convergence speed compared to the GA and PSO.

8 | CONCLUSIONS AND FUTURE PERSPECTIVES

The paper provides an overview of the DSM techniques, sustainability of SG, DSM optimization approaches, and the challenges of the DSM. Therefore, a comprehensive comparison among different algorithms used in DSM optimization problems is presented in terms of various factors, such as reduction of PAR, reduction of energy

TABLE 5 Advantages and disadvantages of various algorithms used in demand side management

Algorithm	Advantages	Disadvantages	References
GA	The GA is stochastic. It can be easily parallelized to solve problems with multiple solutions as it searches from a population of points and not a single point. GA theory is not complex. For this reason, various discrete and continuous mathematical models and simulations can be easily solved by the GA.	It has the tendency to converge toward the local optima or even arbitrary points rather than the global optimum of the problem. Hence, the GA does not know how to sacrifice short-term fitness to gain longer-term fitness. Sometimes, the GA requires expensive fitness function evaluation to find the optimal solution to complex, high-dimensional, multimodal problems.	219-222
PSO	The PSO algorithm is easy to implement, and it solves problems by scattering and optimization. The mathematical calculation of the PSO algorithm is simple. PSO uses heuristic methods with the most successive and maximum efficiency rates. It has a very fast search speed. It optimizes multi-objective problems.	The PSO algorithm easily falls into the local optimum in a high-dimensional space. In the iterative process, PSO has a low convergence rate. Some modification is required because of its conflicting nature.	161,193,219,223-227
ACO	ACO has low computational complexity and low computational time. In the case of optimization, it has good adaptability and positive feedback mechanism.	As a result of its many parameters, ACO is not suitable for solving a complex problem. Its performance can be comparatively lower than that of the GA.	155,228-230
GTA	The GTA is easy to understand. Mathematically, it determines the best strategy for given conditions and optimizes outcomes.	During the solving of games involving mixed strategies, the GTA causes a complicated large pay-off matrix. Sometimes, all the competitive problems cannot be solved and analyzed by the GTA.	34,46,184
LP	The LP technique develops the quality of decisions and achieves the optimum usage of productive resources. Linear problem optimization is considered as competent for analyzing probability index reliability and economic constraints. It provides a promising and practical solution to problems.	The problem represented by an objective function and other constraints must be in linear form, which is not possible.	191,231-234
NLP method	NLP provides simple operation for complex problems.	The NLP method sometimes involves great computational burden from problems. The numerical approach-based NLP works with several iterations.	193,235-237
DP method	It is a simple and efficient algorithm that helps to achieve global optimum. It is suitable for taking decisions on investment for new systems.	This method has no definite formulation. Each problem must be solved according to its own parameters. It has numbers of recursive functions, which may create confusion in the coding.	238

Abbreviations: ACO, ant colony optimization; GTA, game theory algorithm; LP, linear programming; NLP, non-LP; PSO, particle swarm optimization.

cost, and types of DR programs. The DSM is associated with many challenges, such as security and privacy of SG networks; however a number of measures have been adopted by international organizations to overcome these issues. From the detail literature review, it was observed that a single algorithm has a number of disadvantages including a high complexity with respect to constraints, low convergence rate, conflicting nature, and low performance. Therefore, single algorithms may not be the best way to handle the complexity of DSM optimization problems. In that prospect, some modified algorithms and hybrid algorithms can overcome the shortcomings of single algorithms and outperform the single algorithms on the basis of convergence rate, complexity, noisy environment, imprecision, uncertainty, and vagueness. Taking into account the recent advances of DSM in SG, the paper has summarized some suggestions for future research directions as follows:

- Distributed energy generation from renewable resources can play key role for attaining grid sustainability. However, the distributed generation is variable due to the irregular and unpredictable characteristics of RESs. Therefore, optimization of hybrid renewables with their real-time and cost-effective implementation and selection of appropriate storage system require further extensive investigations.
- When implementing a DSM program, fairness among different users, standardization, and interoperability of SG needs to be assured. In order to ensure the security and privacy of customers' data, an in-depth research is required for the real-time synchronization and integration among the security, safety, smart appliances, and monitoring. Apart from this, the agencies, stockholders, and policy makers should come forward to implement new regulations and policies for enhancing the customers' trust.
- The hybrid algorithms-based optimization models improve the convergence and computation time of DSM optimization problems. However, the types of problems (ie, single or multi-objective), optimization types (ie, local or global), robustness, and accuracy should be further considered, when selecting algorithm for solving DSM optimization problems.

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ENDNOTE

¹Peak clipping, valley filling and load shifting are the traditional load management actions.

REFERENCES

1. Halder PK, Parvez MS. Financial analyses and social impacts of solar home systems in Bangladesh: a case study. *Int J Renewable Energy Res.* 2015;5(2):398-403.
2. Khan I, Halder PK. Electrical energy conservation through human behavior change: perspective in Bangladesh. *Int J Renewable Energy Res.* 2016;6(1):43-52.
3. Halder PK, Paul N, Joardder MUH, Sarker M. Energy scarcity and potential of renewable energy in Bangladesh. *Renewable Sustainable Energy Rev.* 2015;51:1636-1649.
4. Caineng Z, Qun Z, Guosheng Z, Bo X. Energy revolution: from a fossil energy era to a new energy era. *Nat Gas Ind B.* 2016;3(1):1-11.
5. Shearer C, Bistline J, Inman M, Davis SJ. The effect of natural gas supply on US renewable energy and CO₂ emissions. *Environ Res Lett.* 2014;9:1-8.
6. Jackson RB, Le QC, Andrew RM, et al. Global energy growth is outpacing decarbonization. *Environ Res Lett.* 2018; 13:1-7.
7. Warren P. A review of demand-side management policy in the UK. *Renewable Sustainable Energy Rev.* 2014;29:941-951.
8. Halder P, Kundu S, Patel S, Ramezani M, Parthasarathy R. A comparison of ionic liquids and organic solvents on the separation of cellulose-rich material from river red gum. *Bioenergy Res.* 2019;12:275-291.
9. Li Q, Zhou M. The future-oriented grid-smart grid. *J Comput.* 2011;6(1):98-105.
10. Agrawal P. Overview of DOE microgrid activities. Paper presented at: Symposium on Microgrids. Montreal; 2006: pp. 1-32.
11. Alizadeh M, Li X, Wang Z, Scaglione A, Melton R. Demand-side management in the smart grid: information processing for the power switch. *IEEE Signal Process Mag.* 2012;29(5): 55-67.
12. Ding K, Zhi J. Wind power peak-valley regulation and frequency control technology. In: Wang K, Kang C, Ren D, ed. *Large-Scale Wind Power Grid Integration*. San Diego, CA: China Electric Power Press. Published by Elsevier Inc; 2016: 211-232.
13. Maharjan S, Zhang Y, Gjessing S, Tsang D. User-centric demand response management in the smart grid with multiple providers. *IEEE Trans Emerg Top Comput.* 2014;5(4):494-505.
14. Gellings CW. Evolving practice of demand-side management. *J Mod Power Syst Clean Energy.* 2017;5(1):1-9.
15. Qureshi WA, Nair N-KC, Farid MM. Impact of energy storage in buildings on electricity demand side management. *Energy Convers Manage.* 2011;52(5):2110-2120.
16. Wu Y, Ravey A, Chrenko D, Miraoui A. Demand side energy management of EV charging stations by approximate dynamic programming. *Energy Convers Manage.* 2019;196:878-890.
17. Wang Y, Yang Z, Mourshed M, Guo Y, Niu Q, Zhu X. Demand side management of plug-in electric vehicles and coordinated unit commitment: a novel parallel competitive swarm optimization method. *Energy Convers Manage.* 2019; 196:935-949.

18. Zhang Y, Peng H. Exploring the direct rebound effect of residential electricity consumption: an empirical study in China. *Appl Energy*. 2017;196:132-141.
19. Mahbubur S, Dulal M. The impact of sources of energy production on globalization: evidence from panel data analysis. *Renewable Sustainable Energy Rev*. 2017;74:110-115.
20. Awais M, Javaid N, Shaheen N, et al. An efficient genetic algorithm based demand side management scheme for smart grid. Paper presented at: 18th International Conference on Network-Based Information Systems. 2015: pp. 351-356.
21. Rahim S, Javaid N, Ahmad A, et al. Exploiting heuristic algorithms to efficiently utilize energy management controllers with renewable energy sources. *Energy Buildings*. 2016;129: 452-470.
22. Zhao Z, Lee WC, Shin Y, Member S, Song K. An optimal power scheduling method for demand response in home energy management system. *IEEE Trans Smart Grid*. 2013; 4(3):1391-1400.
23. Khan MA, Javaid N, Mahmood A, Khan ZA. A generic demand-side management model for smart grid. *Int J Energy Res*. 2015;39:954-964.
24. Arabali A, Ghofrani M, Fadali MS, Baghzouz Y. Genetic-algorithm-based optimization approach for energy management. *IEEE Trans Power Deliv*. 2013;28(1):162-170.
25. Hussain H, Javaid N, Iqbal S, Hasan Q, Aurangzeb K, Alhussein M. An efficient demand side management system with a new optimized home energy management controller in smart grid. *Energies*. 2018;11(1):1-28.
26. Kakran S, Chanana S. Smart operations of smart grids integrated with distributed generation: a review. *Renewable Sustainable Energy Rev*. 2018;81:524-535.
27. Lu X, Zhou K, Zhang X, Yang S. A systematic review of supply and demand side optimal load scheduling in a smart grid environment. *J Clean Prod*. 2018;203:757-768.
28. Fang X, Misra S, Xue G, Yang D. Smart grid—the new and improved power grid: a survey. *IEEE Commun Surv Tutor*. 2012;14(4):944-980.
29. Farhangi H. The path of the smart grid. *IEEE Power Energy Mag*. 2010;8(1):18-28.
30. Zhong J, Zheng R, Yang W, Wu F. Construction of smart grid at information age. *Power Syst Technol*. 2009;33(13):12-18.
31. Yi T, Pipattanasomporn M S-n S, Hao-ming L, Rahman S. Comparative study on smart grid related R&D in China, the United States and the European Union. *Power Syst Technol*. 2009;33(15):7-15.
32. Kolokotsa D. The role of smart grids in the building sector. *Energy Buildings*. 2016;116:703-708.
33. Macedo M, Galo J, Almeida L, Lima A. Opportunities and challenges of DSM in smart grid environment. Paper presented at: Third International Conference on Smart Grids, Green Communications and IT Energy-Aware Technologies. 2013: pp. 156-160.
34. Nguyen H, Song J, Han Z. Demand side management to reduce peak-to-average ratio using game theory in smart grid. Paper presented at: IEEE Conference on Computer Communications Workshops. 2012: pp. 91-96.
35. Zhang N, Hu Z, Dai D, Dang S, Yao M, Zhou Y. Unit commitment model in smart grid environment considering carbon emissions trading. *IEEE Trans Smart Grid*. 2015;7(1):420-427.
36. Xin A, Xiao L, Wen Jie Q, Yang W. Bid-scheduling of demand side reserve based on demand response considering carbon emission trading in smart grid. Paper presented at: 5th International Conference on Critical Infrastructure. 2010: pp. 1-6.
37. Li D, Sun H, Chiu W-Y. Achieving low carbon emission using smart grid technologies. Paper presented at: IEEE 85th Vehicular Technology Conference: pp. 1-5.
38. Gellings CW, Chamberlin JH. *Demand-side Management: Concepts and Methods*. Lilburn, Georgia: Fairmont Press; 1987.
39. Yang H, Zhang Y, Tong X. System dynamics model for demand side management. Paper presented at: 3rd International Conference on Electrical and Electronics Engineering. 2006: pp. 1-4.
40. Logenthiran T, Srinivasan D, Shun TZ. Demand side management in smart grid using heuristic optimization. *IEEE Trans Smart Grid*. 2012;3(3):1244-1252.
41. Kinhekar N, Padhy NP, Gupta HO. Multiobjective demand side management solutions for utilities with peak demand deficit. *Int J Electr Power Energy Syst*. 2014;55:612-619.
42. Gelazanskas L, Gamage KAA. Demand side management in smart grid: a review and proposals for future direction. *Sustainable Cities Soc*. 2014;11:22-30.
43. Gellings CW. The concept of demand-side management for electric utilities. *Proc IEEE*. 1985;73(10):1468-1470.
44. Sinha A, De M. Load shifting technique for reduction of peak generation capacity requirement in smart grid. Paper presented at: IEEE International Conference on Power Electronics, Intelligent Control and Energy Systems. 2016: pp. 1-5.
45. Raza A, Abas N, Ul Q, Seyedmahmoudian M. Demand side management in hybrid rooftop photovoltaic integrated smart nano grid. *J Clean Prod*. 2020;258:120747.
46. Song L, Xiao Y, Van Der Schaar M. Demand side management in smart grids using a repeated game framework. *IEEE J Sel Areas Commun*. 2014;32(7):1412-1424.
47. Ruiz N, Cobelo I, Oyarzabal J. A direct load control model for virtual power plant management. *IEEE Trans Power Syst*. 2009;24(2):959-966.
48. Yao L, Member S, Chang W, Yen R. An iterative deepening genetic algorithm for scheduling of direct load control. *IEEE Trans Power Syst*. 2005;20(3):1414-1421.
49. Kothari DP, Nagrath IJ. *Modern Power System Analysis*. New Delhi, India: McGraw-Hill Higher Education; 2008.
50. Pina A, Silva C, Ferrão P. The impact of demand side management strategies in the penetration of renewable electricity. *Energy*. 2012;41(1):128-137.
51. Infield DG, Member S, Short J, Horne C, Freris LL. Potential for domestic dynamic demand-side management in the UK. Paper presented at: IEEE Power Engineering Society General Meeting. 2007: pp. 1-6.
52. Bayod-Rújula AA. Future development of the electricity systems with distributed generation. *Energy*. 2009;34:377-383.
53. Torriti J, Hassan MG, Leach M. Demand response experience in Europe: policies, programmes and implementation. *Energy*. 2010;35:1575-1583.
54. Wang J, Bloyd CN, Hu Z, Tan Z. Demand response in China. *Energy*. 2010;35(4):1592-1597.
55. Palensky P, Dietrich D. Demand side management: demand response, intelligent energy systems, and smart loads. *IEEE Trans Ind Inf*. 2011;7(3):381-388.

56. Alasser R, Tripathi A, Rao TJ, Sreekanth KJ. A review on implementation strategies for demand side management (DSM) in Kuwait through incentive-based demand response programs. *Renewable Sustainable Energy Rev.* 2017;77:617-635.
57. Jordehi AR. Optimisation of demand response in electric power systems: a review. *Renewable Sustainable Energy Rev.* 2019;103:308-319.
58. Gellings CW. *The Smart Grid: Enabling Energy Efficiency and Demand Response*. Lilburn, Georgia: Fairmont Press; 2009.
59. Logenthiran T, Srinivasan D, Vanessa KWM. Demand side management of smart grid: load shifting and incentives. *J Renewable Sustainable Energy.* 2014;6(3):1-18.
60. Yang P, Chavali P, Gilboa E, Nehorai A. Parallel load schedule optimization with renewable distributed generators in smart grids. *IEEE Trans Smart Grid.* 2013;4(3):1431-1441.
61. Qdr QJ. Benefits of demand response in electricity markets and recommendations for achieving them. Washington. 2006.
62. Siano P. Demand response and smart grids - a survey. *Renewable Sustainable Energy Rev.* 2014;30:461-478.
63. Shafie-khah M, Heydarian-forushani E, Member S, et al. Optimal behavior of electric vehicle parking lots as demand response aggregation agents. *IEEE Trans Smart Grid.* 2015;7(6):2654-2665.
64. Rebours Y, Kirschen DS. *What Is Spinning Reserve?*. Manchester, England: The University of Manchester; 2005.
65. Ortega-vazquez MA, Kirschen DS. Estimating the spinning reserve requirements in systems with significant wind power generation penetration. *IEEE Trans Power Syst.* 2009;24(1):114-124.
66. Tuballa ML, Abundo ML. A review of the development of smart grid technologies. *Renewable Sustainable Energy Rev.* 2016;59:710-725.
67. Metke AR, Ekl RL. Security technology for smart grid networks. *IEEE Trans Smart Grid.* 2010;1(1):99-107.
68. Van Dievel P, De Vos K, Belmans R. Demand response in electricity distribution grids: regulatory framework and barriers. Paper presented at: Int Conf Eur Energy Mark EEM 2014; pp. 1-5.
69. Kabalci Y. A survey on smart metering and smart grid communication. *Renewable Sustainable Energy Rev.* 2016;57:302-318.
70. Baharlouei Z, Hashemi M. Demand side management challenges in smart grid: a review. Paper presented at: Smart Grid Conference. 2013; pp. 96-101.
71. McDaniel P, McLaughlin S. Security and privacy challenges in the smart grid. *IEEE Secur Priv.* 2009;7(3):75-77.
72. Dawoud B, Amer E, Gross D. Experimental investigation of an adsorptive thermal energy storage. *Int J Energy Res.* 2007;31:135-147.
73. Parikh PP, Kanabar MG, Sidhu TS. Opportunities and challenges of wireless communication technologies for smart grid applications. Paper presented at: IEEE PES Gen Meet 2010; pp. 1-7.
74. Fan Z, Kulkarni P, Gormus S, et al. Smart grid communications: overview of research challenges, solutions, and standardization activities. *IEEE Commun Surv Tutorials.* 2013;15(1):21-38.
75. Souryal MR, Golmie N. Analysis of advanced metering over a wide area cellular network. Paper presented at: 2011 IEEE Int Conf Smart Grid Commun. 2011; pp. 102-107.
76. Iyer G, Agrawal P, Monnerie E, Cardozo RS. Performance analysis of wireless mesh routing protocols for smart utility networks. 2011 IEEE Int Conf Smart Grid Commun 2011; pp. 114-119.
77. Kuzlu M, Pipattanasompom M, Rahman S. A comprehensive review of smart grid related standards and protocols. Paper presented at: ICSG 2017 - 5th Int Istanbul Smart Grids Cities Congr Fair. 2017; pp. 12-16.
78. MacKiewicz RE. Overview of IEC 61850 and benefits. 2006 IEEE PES Power Syst Conf Expo PSCE 2006 - Proc. 2006; Vol. 57, no. 57; pp. 623-630.
79. IPE Society. *IEEE Standard for SCADA and Automation Systems*. Vol 2008. New York, NY: IEEE; 2008.
80. Basso TS, DeBlasio R. IEEE 1547 series of standards: interconnection issues. *IEEE Trans Power Electron.* 2004;19(5):1159-1162.
81. The Advanced Security Acceleration Project (ASAP-SG). Security profile for advanced metering infrastructure. 2012.
82. Cleveland F. IEC TC57 security standards for the power system's information infrastructure - beyond simple encryption. Paper presented at: Proc IEEE Power Eng Soc Transm Distrib Conf 2006; pp. 1079-1087.
83. Khalid H, Amin FR, Chen C. Comparing the status and challenges of demand-side management (DSM) implementation in Asia-Pacific region: a case study of China's power sector. *Energy Procedia.* 2018;152:101-108.
84. Oconnell N, Pinson P, Madsen H, Omalley M. Benefits and challenges of electrical demand response: a critical review. *Renewable Sustainable Energy Rev.* 2014;39:686-699.
85. Vuppala SK, Member S, Padmanabh K, Bose SK. Incorporating fairness within demand response programs in smart grid. Paper presented at: Innovative Smart Grid Technologies 2011; pp. 1-9.
86. Kefayati M, Baldick R. Energy delivery transaction pricing for flexible electrical loads. Paper presented at: IEEE International Conference on Smart Grid Communication. 2011; pp. 363-368.
87. Baharlouei Z, Narimani H, Mohsenian-rad H. Tackling co-existence and fairness challenges in autonomous demand side management. Paper presented at: Global Communications Conference. 2012; 3159-3164.
88. Shinwari M, Youssef A, Hamouda W. A water-filling based scheduling algorithm for the smart grid. *IEEE Trans Smart Grid.* 2012;3(2):710-719.
89. Baharlouei Z, Hashemi M, Narimani H, Mohsenian-rad H. Achieving optimality and fairness in autonomous demand response: benchmarks and billing mechanisms. *IEEE Trans Smart Grid.* 2013;4(2):968-975.
90. Güngör VC, Sahin D, Kocak T, et al. Smart grid technologies: communication technologies and standards. *IEEE Trans Ind Inf.* 2011;7(4):529-539.
91. Hossain E, Han Z, Poor HV. *Smart Grid Communications and Networking*. Cambridge, England: Cambridge University Press; 2012.
92. Office of Electricity/Department of Energy. Retrieved on 28 August 2018 from <https://www.energy.gov/oe/office-electricity>
93. Yan Y, Qian Y, Sharif H, Tipper D. A survey on smart grid communication infrastructures: motivations, requirements and challenges. *IEEE Commun Surv Tutorials.* 2013;15(1):5-20.

94. Erol-Kantarci M, Mouftah HT. Wireless multimedia sensor and actor networks for the next generation power grid. *Ad Hoc Networks*. 2011;9(4):542-551.
95. Gabash A, Li P. Flexible optimal operation of battery storage systems for energy supply networks. *IEEE Trans Power Syst*. 2013;28(3):2788-2797.
96. Kanchev H, Lu D, Colas F, Lazarov V, Francois B, Member S. Energy management and operational planning of a microgrid with a PV-based active generator for smart grid applications. *IEEE Trans Ind Electron*. 2011;58(10):4583-4592.
97. Wu Z, Tazvinga H, Xia X. Demand side management of photovoltaic-battery hybrid system. *Appl Energy*. 2015;148:294-304.
98. Blake ST, Sullivan DTJO. Optimization of distributed energy resources in an industrial microgrid. *Procedia CIRP*. 2018;67:104-109.
99. Quiggin D, Cornell S, Tierney M, Buswell R. A simulation and optimisation study: towards a decentralised microgrid, using real world fluctuation data. *Energy*. 2012;41(1):549-559.
100. Palma-Behnke R, Benavides C, Aranda E, Llanos J, Sáez D. Energy management system for a renewable based microgrid with a demand side management mechanism. Paper presented at: IEEE Symposium on Computational Intelligence Applications In Smart Grid. 2011: pp. 1-8.
101. Fan W, Liu N, Zhang J. An event-triggered online energy management algorithm of smart home: Lyapunov optimization approach. *Energies*. 2016;9(5):1-24.
102. Yang P, Chavali P, Nehorai A. Parallel autonomous optimization of demand response with renewable distributed generators. Paper presented at: 2012 IEEE 3rd Int Conf Smart Grid Commun. 2012: pp. 55-60.
103. Shen J, Jiang C, Liu Y, Qian J. A microgrid energy management system with demand response for providing grid peak shaving. *Electr Power Components Syst*. 2016;44(8):843-852.
104. Solanki BV, Raghurajan A, Bhattacharya K, Canizares CA. Including smart loads for optimal demand response in integrated energy management systems for isolated microgrids. *IEEE Trans Smart Grid*. 2017;8(4):1739-1748.
105. Dietrich K, Latorre JM, Olmos L, Ramos A. Demand response in an isolated system with high wind integration. *IEEE Trans Power Syst*. 2012;27(1):20-29.
106. Nguyen HK, Bin SJ, Han Z. Distributed demand side management with energy storage in smart grid. *IEEE Trans Parallel Distrib Syst*. 2015;26(12):3346-3357.
107. Aghajani GR, Shayanfar HA, Shayeghi H. Demand side management in a smart micro-grid in the presence of renewable generation and demand response. *Energy*. 2017;126:622-637.
108. Çiçek N, Deliç H. Demand response management for smart grids with wind power. *IEEE Trans Sustainable Energy*. 2015;6(2):625-634.
109. Amrollahi MH, Bathaee SMT. Techno-economic optimization of hybrid photovoltaic/wind generation together with energy storage system in a stand-alone micro-grid subjected to demand response. *Appl Energy*. 2017;202:66-77.
110. Wang X, Palazoglu A, El-Farra NH. Operational optimization and demand response of hybrid renewable energy systems. *Appl Energy*. 2015;143:324-335.
111. Philippou N, Venizelou V, Makrides G, Hadjipanayi M, Efthymiou V, Georgiou GE. PV integration and price-based Demand Side Management: optimum time-of-use tariffs. Paper presented at: Proc 33th EUPVSEC. 2016.
112. Behboodi S, Chassin DP, Djilali N, Crawford C. Interconnection-wide hour-ahead scheduling in the presence of intermittent renewables and demand response: a surplus maximizing approach. *Appl Energy*. 2017;189:336-351.
113. Das CK, Bass O, Kothapalli G, Mahmoud TS, Habibi D. Overview of energy storage systems in distribution networks: placement, sizing, operation, and power quality. *Renewable Sustainable Energy Rev*. 2019;91:1205-1230.
114. Abbott M, Cohen B. Issues associated with the possible contribution of battery energy storage in ensuring a stable electricity system. *Electr J*. 2020;33(6):106771.
115. Ajanovic A, Hiesl A, Haas R. On the role of storage for electricity in smart energy systems. *Energy*. 2020;200:117473.
116. Alagoz BB, Kaygusuz A, Karabiber A. A user-mode distributed energy management architecture for smart grid applications. *Energy*. 2012;44:167-177.
117. Longe OM, Ouahada K, Rimer S, Harutyunyan AN, Ferreira HC. Distributed demand side management with battery storage for smart home energy scheduling. *Sustainability*. 2017;9(1):1-13.
118. Nguyen HK, Bin SJ, Han Z. Distributed demand side management with energy storage in smart grid storage in smart grid. *IEEE Trans Parallel Distrib Syst*. 2015;26(12):3346-3357.
119. Pilz M, Al-Fagih L. A dynamic game approach for demand-side management: scheduling energy storage with forecasting errors. *Dyn Games Appl*. 2019;1-33.
120. Ozdemir E. Dynamic energy management for photovoltaic power system including hybrid energy storage in smart grid applications. *Energy*. 2018;162:72-82.
121. Shen L, Cheng Q, Cheng Y, Wei L, Wang Y. Hierarchical control of DC micro-grid for photovoltaic EV charging station based on flywheel and battery energy storage system. *Electr Power Syst Res*. 2020;179:106079.
122. Man KF, Tang KS, Kwong S. Genetic algorithms: concepts and applications. *IEEE Trans Ind Electron*. 1996;43(5):519-534.
123. Bozorg-Haddad O, Solgi M, Loáiciga HA. *Meta-Heuristic and Evolutionary Algorithms for Engineering Optimization*. Hoboken, NJ: John Wiley & Sons, Inc.; 2017.
124. Back T. Selective pressure in evolutionary algorithms: a characterization of selection mechanisms. Paper presented at: IEEE conference on evolutionary computation. IEEE World Congress on Computational Intelligence. 1994: pp. 57-62.
125. Appl J, Valenzuela F. Scheduling electric power generators using particle swarm optimization combined with the Lagrangian relaxation method. *Int J Appl Math Comput Sci*. 2004;14(3):411-421.
126. Chuanwen J, Bompard E. A self-adaptive chaotic particle swarm algorithm for short term hydroelectric system scheduling in deregulated environment. *Energy Convers Manage*. 2005;46(17):2689-2696.
127. Gaur S, Ch S, Graillot D, Chahar BR, Kumar DN. Application of artificial neural networks and particle swarm optimization for the management of groundwater resources. *Water Resour Manag*. 2013;27(3):927-941.
128. Saadatpour M, Afshar A. Multiobjective simulation-optimization approach in pollution spill response management model in reservoirs. *Water Resour Manag*. 2013;27(6):1851-1865.

129. Afshar A, Massoumi F, Afshar A, Mariño MA. State of the art review of ant colony optimization applications in water resource management. *Water Resour Manag*. 2015;29(11):3891-3904.
130. Christodoulou S. Scheduling resource-constrained projects with ant colony optimization artificial agents. *J Comput Civ Eng*. 2009;24(1):45-55.
131. Merkle D, Middendorf M, Schmeck H. Ant colony optimization for resource-constrained project scheduling. *IEEE Trans Evol Comput*. 2002;6(4):333-346.
132. Dorigo M, Maniezzo V, Coloni A. Ant system: optimization by a colony of cooperating agents. *IEEE Trans Syst Man, Cybern Part B Cybern*. 1996;26(1):29-41.
133. Vanderbei RJ. *Linear Programming: Foundations and Extensions*. Boston, MA: Springer; 1997.
134. Karlin S. *Matrix Games, Programming, and Mathematical Economics: Mathematical Methods and Theory in Games, Programming, and Economics*. California, CA: Addison-Wesley Publishing Company, Elsevier Inc; 1959.
135. Bitran GR, Novaes AG. Linear programming with a fractional objective function. *Oper Res*. 1973;21:22-29.
136. Hillier FS, Lieberman GJ. *Introduction to Operations Research*. Boston, NY: McGraw-Hill Higher Education; 2010.
137. Gill PE, Murray W, Saunders MA, Wright MH. Constrained nonlinear programming. *Handbooks in Operations Research and Management Science*. North-Holland: Elsevier Science Publishers B.V; 1989:171-210.
138. Zangwill WI. *Nonlinear Programming: a Unified Approach*. Englewood Cliffs, NJ: Prentice-Hall; 1969.
139. Vanderbei RJ, Shanno DF. An interior-point method for non-convex nonlinear programming. *Comput Optim Appl*. 1999;13: 231-252.
140. Hadley GF. Nonlinear and dynamic programming. *Econometrica*. 1966;34(1):248-249.
141. Bellman RE, Dreyfus SE. *Applied Dynamic Programming*. Princeton, NJ: Princeton University Press; 2015.
142. Beasley D, Bull DR, Martin RR. An overview of genetic algorithms: part 1, fundamentals. *Univ Comput*. 1993;15(2):56-69.
143. Bourouni K, Ben M'Barek T, Al Taei A. Design and optimization of desalination reverse osmosis plants driven by renewable energies using genetic algorithms. *Renewable Energy*. 2011;36(3):936-950.
144. Yang H, Zhou W, Lu L, Fang Z. Optimal sizing method for stand-alone hybrid solar-wind system with LPSP technology by using genetic algorithm. *Sol Energy*. 2008;82(4):354-367.
145. Mitchell M. Handbook of genetic algorithms. *Artif Intell*. 1998; 100(1):325-330.
146. Michalewicz Z. Evolution strategies and other methods. *Genetic Algorithms, Data Structures and Evolution Programs*. New York, NY: Springer; 1996:159-177.
147. Kang CC, Chuang YJ, Tung KC, et al. A genetic algorithm-based boolean delay model of intracellular signal transduction in inflammation. *BMC Bioinf*. 2011;12(1):10-12.
148. Mohsenian-Rad AH, Wong VWS, Jatskevich J, Schober R. Optimal and autonomous incentive-based energy consumption scheduling algorithm for smart grid. Paper presented at: Innovative Smart Grid Technologies Conference 2010: pp. 1-6.
149. Wang K, Li H, Maharjan S, Zhang Y, Guo S. Green energy scheduling for demand side management in the smart grid. *IEEE Trans Green Commun Network*. 2018;2(2):596-611.
150. Logenthiran T, Srinivasan D, Phyu E. Particle swarm optimization for demand side management in smart grid. Paper presented at: IEEE Innovative Smart Grid Technologies-Asia. 2015: pp. 1-6.
151. Samadi P, Mohsenian-Rad H, Schober R, Wong VWS. Advanced demand side management for the future smart grid using mechanism design. *IEEE Trans Smart Grid*. 2012;3(3): 1170-1180.
152. Hyoungchel Bae, Jongha Yoon, Yunseong Lee, et al. User-friendly demand side management for smart grid networks. Paper presented at: The International Conference on Information Networking. 2014: pp. 481-485.
153. Mohsenian-Rad AH, Wong VWS, Jatskevich J, Schober R, Leon-Garcia A. Autonomous demand-side management based on game-theoretic energy consumption scheduling for the future smart grid. *IEEE Trans Smart Grid*. 2010;1(3):320-331.
154. Wang J, Sun Z, Zhou Y, Dai J. Optimal dispatching model of smart home energy management system. Paper presented at: IEEE Innovative Smart Grid Technologies-Asia. 2012: pp. 1-5.
155. Liu B, Kang J, Jiang N, Jing Y. Cost control of the transmission congestion management in electricity systems based on ant colony algorithm. *Energy Power Eng*. 2011;03(01):17-23.
156. Mahmood A, Ullah MN, Razzaq S, et al. A new scheme for demand side management in future smart grid networks. *Procedia Comput Sci*. 2014;32:477-484.
157. Bharathi C, Rekha D, Vijayakumar V. Genetic algorithm based demand side management for smart grid. *Wirel Pers Commun*. 2017;93(2):481-502.
158. Jayadev V, Swarup KS. Optimization of microgrid with demand side management using genetic algorithm Indian Institute of Technology Madras, India. Paper presented at: IET Conference on Power in Unity: A Whole System Approach. 2013: pp. 1-12.
159. Kennedy J, Eberhart R. Particle swarm optimization. Paper presented at: International Conference on Neural Networks. 1995: pp. 1942-1948.
160. Eberhart R, Kennedy J. A new optimizer using particle swarm theory. Paper presented at: Sixth International Symposium on Micro Machine and Human Science. 1995: pp. 39-43.
161. Amer M, Namaane A, Sirdi NKM. Optimization of hybrid renewable energy systems (HRES) using PSO for cost reduction. *Energy Procedia*. 2013;42:318-327.
162. Armaghani DJ, Hajihassani M, Mohamad ET, Marto A, Noorani SA. Blasting-induced flyrock and ground vibration prediction through an expert artificial neural network based on particle swarm optimization. *Arab J Geosci*. 2014;7(12): 5383-5396.
163. Connor J, Seyedmahmoudian M, Horan B. Using particle swarm optimization for PID optimization for altitude control on a quadrotor. Paper presented at: Australasian Universities Power Engineering Conference. 2018: 1-6.
164. Nayak SK, Sahoo NC, Panda G. Demand side management of residential loads in a smart grid using 2D particle swarm optimization technique. Paper presented at: IEEE Power, Communication and Information Technology Conference. 2015: pp. 201-206.
165. Kennedy J, Eberhart RC. A discrete binary version of the particle swarm algorithm. Paper presented at: IEEE International Conference on Computational Cybernetics and Simulation. 1997: pp. 4104-4108.

166. Pedrasa MAA, Member S, Spooner TD, Macgill IF. Scheduling of demand side resources using binary particle swarm optimization. *IEEE Trans Power Syst.* 2009;24(3):1173-1181.
167. Mahmood D, Javaid N, Alrajeh N, et al. Realistic scheduling mechanism for smart homes. *Energies.* 2016;9(3):1-28.
168. Zhou Y, Chen Y, Xu G, Zhang Q, Krundel L. Home energy management with PSO in smart grid. Paper presented at: IEEE 23rd International Symposium Industrial Electronics. 2014: pp. 1666-1670.
169. Zhou Y, Xu G. Demand side energy management with PSO and regulated electric vehicles behaviours. Paper presented at: IEEE PES Asia-Pacific Power and Energy Engineering Conference. 2014: pp. 1-6.
170. Dorigo M, Gambardella LM. Ant colony system: a cooperative learning approach to the traveling salesman problem. *IEEE Trans Evol Comput.* 1997;1(1):53-66.
171. Dorigo M, Birattari M, Stutzle T. Ant colony optimization. *IEEE Comput Intell Mag.* 2006;1(4):28-39.
172. Dethlefs T, Preisler T, Renz W. Ant-colony based self-optimization for demand-side-management. Paper presented at: Smarter Europe. 2015: pp. 1-8.
173. Okonta CI, Kemp AH, Edopkia RO, Monyei GC, Okelue ED. A heuristic based ant colony optimization algorithm for energy efficient smart homes. Paper presented at: 5th International Conference & Exhibition on Clean Energy. 2016: pp. 1-12.
174. Rahim S, Iqbal Z, Shaheen N, et al. Ant colony optimization based energy management controller for smart grid. Paper presented at: IEEE 30th International Conference on Advanced Information Networking and Applications. 2016: pp. 1154-1159.
175. Hazra J, Das K, Seetharam DP. Smart grid congestion management through demand response. Paper presented at: IEEE Third International Conference on Smart Grid Communications. 2012: pp. 109-114.
176. Deng R, Yang Z, Chow MY, Chen J. A survey on demand response in smart grids: mathematical models and approaches. *IEEE Trans Ind Inf.* 2015;11(3):570-582.
177. Bu S, Yu FR. A game-theoretical scheme in the smart grid with demand-side management: towards a smart cyber-physical power infrastructure. *IEEE Trans Emerg Top Comput.* 2013;1(1):22-32.
178. Yang P, Tang G, Nehorai A. A game-theoretic approach for optimal time-of-use electricity pricing. *IEEE Trans Power Syst.* 2013;28(2):884-892.
179. Saad W, Han Z, Poor HV, Başar T. Game-theoretic methods for the smart grid: an overview of microgrid systems, demand-side management, and smart grid communications. *IEEE Signal Process Mag.* 2012;29(5):86-105.
180. Chen HH, Member S, Li Y, Member S. Autonomous demand side management with instantaneous load billing: an aggregative game approach. *IEEE Trans Smart Grid.* 2014;5(4):1744-1754.
181. Smets B. Investigation on demand side management techniques in the smart grid using game theory and ICT concepts. University of Antwerp. *arXiv preprint.* 2014.
182. Sofana Reka S, Ramesh V. A demand response modeling for residential consumers in smart grid environment using game theory based energy scheduling algorithm. *Ain Shams Eng J.* 2016;7(2):835-845.
183. Kapade N, Deshmukh S. Comparative study of game theoretic approaches in distributed system. Paper presented at: World Congress on Computer and Information Technology. 2013: pp. 1-4.
184. Deng R, Member S, Yang Z, Chen J, Member S. Residential energy consumption scheduling: a coupled-constraint game approach. *IEEE Trans Smart Grid.* 2014;5(3):1340-1350.
185. Dantzig GB, Thapa MN. *Linear Programming 1.* New York, NY: Springer; 1997.
186. Schäffler S. *Global Optimization: A Stochastic Approach.* New York, NY: Springer; 2012.
187. Bradley J. Optimal pivot point selection for the simplex method. 2012.
188. Sheblt GB. Direct load control-a profit-based load management using linear programming. *IEEE Trans Power Syst.* 1998; 13(2):688-695.
189. Kurucz CN, Brandt D, Sim S. Linear programming model for reducing system peak through customer load control programs. *IEEE Trans Power Syst.* 1996;11(4):1817-1824.
190. Zhu Z, Tang J, Lambbotharan S, Chin WH, Fan Z. An integer linear programming based optimization for home demand-side management in smart grid. Paper presented at: IEEE PES Innovative Smart Grid Technologies 2012: pp. 1-5.
191. Martins AG, Coelho D, Antunes CH, Clímaco J. A multiple objective linear programming approach to power generation planning with demand-side management (DSM). *Int Trans Oper Res.* 1996;3(3-4):305-317.
192. Avriel M. *Nonlinear Programming: Analysis and Methods.* Englewood Cliffs, NJ: Dover Publications Inc; 2003.
193. Ghofrani M, Hosseini NN. Optimizing hybrid renewable energy systems: a review. *Sustainable Energy-Technological Issues, Applications and Case Studies.* Rijeka, Croatia: InTech; 2016:161-176.
194. Hillier L. *Nonlinear programming.* University of Iowa. 2018.
195. James M. Computational tools for smart grid design. In: Hanzo L, ed. *Smart Grid: Fundamentals of Design and Analysis.* Piscataway, NJ: Wiley; 2012.
196. Kontorinaki M, Spiliopoulou A, Papamichail I, Papageorgiou M, Tyrinopoulos Y, Chrysoulakis J. Overview of nonlinear programming methods suitable for calibration of traffic flow models. *Oper Res.* 2015;15(3):327-336.
197. Shaaban MF, Osman AH, Hassan MS. Day-ahead optimal scheduling for demand side management in smart grids. Paper presented at: European Modelling Symposium 2016: pp. 124-129.
198. Helal SA, Najee RJ, Hanna MO, Shaaban MF, Osman AH, Hassan MS. An energy management system for hybrid microgrids in remote communities. *IEEE 30th Can Conf Electr Comput Eng.* 2017:1-4.
199. Bellman R. *Dynamic Programming.* Princeton, NJ: Dover Publications Inc; 2003.
200. Bertsekas DP. *Dynamic Programming and Optimal Control.* Belmont, MA: Athena Scientific; 2018.
201. Chu WC, Chen BK, Fu CK. Scheduling of direct load control to minimize load reduction for a utility suffering from generation shortage. *IEEE Trans Power Syst.* 1993;8(4):1525-1530.
202. Cohen AI, Wang CC. An optimization method for load management scheduling. *IEEE Trans Power Syst.* 1988;3(2): 612-618.

203. Hsu Y, Member S. Dispatch of direct load control using dynamic programming. *IEEE Trans Power Syst.* 1991;6(3): 1056-1061.
204. Reka SS, Ramesh V. Demand side management scheme in smart grid with cloud computing approach using stochastic dynamic programming. *Perspect Sci.* 2016;8:169-171.
205. Esther BP, Krishna KS, Kumar KS, Ravi K. Demand side management using bacterial foraging optimization algorithm. *Information Systems Design and Intelligent Applications.* New Delhi, India: Springer; 2016;657-666.
206. Barolli L, Woungang I, Hussain OK. *Advances in Intelligent Networking and Collaborative Systems.* Cham, Switzerland: Springer; 2018.
207. Zafar A, Shah S, Khalid R, Hussain SM, Rahim H, Javaid N. A meta-heuristic home energy management system. Paper presented at: IEEE International Conference on Advanced Information Networking and Applications Workshops. 2017: pp. 244-250.
208. Rasheed M, Javaid N, Ahmad A, Khan Z, Qasim U, Alrajeh N. An efficient power scheduling scheme for residential load management in smart homes. *Appl Sci.* 2015;5(4): 1134-1163.
209. Wu B, Ma H, Pan Z, Wang J, Qu W, Wang B. Drying and quality characteristics and models of carrot slices under catalytic infrared heating. *Int Agric Eng J.* 2014;23(2):70-79.
210. Wang L, Wang Z, Yang R. Intelligent multiagent control system for energy and comfort management in smart and sustainable buildings. *IEEE Trans Smart Grid.* 2012;3(2):605-617.
211. Ru N, Jianhua Y. A GA and particle swarm optimization based hybrid algorithm. Paper presented at: IEEE Congress on Evolutionary Computation 2008: pp. 1047-1050.
212. Javaid N, Javaid S, Abdul W, et al. A hybrid genetic wind driven heuristic optimization algorithm for demand side management in smart grid. *Energies.* 2017;10(3):1-27.
213. Ahmad A, Khan A, Javaid N, et al. An optimized home energy management system. *Energies.* 2017;10:1-35.
214. Yang HT, Yang CT, Tsai CC, Chen GJ, Chen SY. Improved PSO based home energy management systems integrated with demand response in a smart grid. Paper presented at: 2015 IEEE Congress on Evolutionary Computation, CEC 2015-Proceedings. 2015: pp. 275-282.
215. Manzoor A, Javaid N, Ullah I, Abdul W, Almogren A, Alamri A. An intelligent hybrid heuristic scheme for smart metering based demand side management in smart homes. *Energies.* 2017;10(9):1-28.
216. Zhang J, Wu Y, Guo Y, Wang B, Wang H, Liu H. A hybrid harmony search algorithm with differential evolution for day-ahead scheduling problem of a microgrid with consideration of power flow constraints. *Appl Energy.* 2016;183:791-804.
217. Pamir, Javaid N, Mohsin SM, Iqbal A, Yasmeen A, Ali I. A Hybrid Bat-Crow Search Algorithm Based Home Energy Management in Smart Grid. In: Barolli L, Javaid N, Ikeda M, Takizawa M, ed. *Complex, Intelligent, and Software Intensive Systems.* Vol. Cham, Switzerland: Springer International Publishing; 2019:75-88.
218. Logenthiran T, Srinivasan D, Khambadkone AM. Multi-agent system for energy resource scheduling of integrated microgrids in a distributed system. *Electr Pow Syst Res.* 2011; 81(1):138-148.
219. Anoune K, Bouya M, Astito A, Ben AA. Sizing methods and optimization techniques for PV-wind based hybrid renewable energy system: a review. *Renewable Sustainable Energy Rev.* 2018;93:652-673.
220. Koutroulis E, Kolokotsa D, Potirakis A, Kalaitzakis K. Methodology for optimal sizing of stand-alone photovoltaic/wind-generator systems using genetic algorithms. *Sol Energy.* 2006; 80:1072-1088.
221. Koutroulis E, Kolokotsa D. Design optimization of desalination systems power-supplied by PV and W / G energy sources. *Desalination.* 2010;258:171-181.
222. Zeng J, Li M, Liu JF, Wu J, Ngan HW. Operational optimization of a stand-alone hybrid renewable energy generation system based on an improved genetic algorithm. Paper presented at: Power and Energy Society General Meeting 2010: pp. 1-6.
223. Li M, Du W, Nian F. An adaptive particle swarm optimization algorithm based on directed weighted complex network. *Math Probl Eng.* 2014;2014:1-7.
224. Hakimi SM. Optimal sizing of a stand-alone hybrid power system via particle swarm optimization for Kahnouj area in south-east of Iran. *Renewable Energy.* 2009;34(7):1855-1862.
225. Lee TY, Chen CL. Wind-photovoltaic capacity coordination for a time-of-use rate industrial user. *IET Renewable Power Gener.* 2009;3(2):152-167.
226. Ardakani FJ, Riahy G, Abedi M. Design of an optimum hybrid renewable energy system considering reliability indices. Paper presented at: 18th Iranian Conference on Electrical Engineering. 2010: pp. 842-847.
227. Askarzadeh A, Coelho S. A novel framework for optimization of a grid independent hybrid renewable energy system: a case study of Iran. *Sol Energy.* 2015;112:383-396.
228. Maringer D. *Heuristic optimization. Portfolio Management with Heuristic Optimization.* Boston, MA: Springer-Verlag; 2005:38-76.
229. Adubi SA, Misra S. A comparative study on the ant colony optimization algorithms. Paper presented at: 11th International Conference on Electronics, Computer and Computation. 2014: pp. 1-4.
230. Erdinc O, Uzunoglu M. Optimum design of hybrid renewable energy systems: overview of different approaches. *Renewable Sustainable Energy Rev.* 2012;16(3):1412-1425.
231. Nagabhushana AC, Jyoti R, Raju AB. Economic analysis and comparison of proposed HRES for stand-alone applications at various places in Karnataka state. Paper presented at: IEEE PES International Conference on Innovative Smart Grid Technologies. 2011: pp. 380-385.
232. Almunif A, Fan L. Mixed integer linear programming and nonlinear programming for optimal PMU placement. Paper presented at: North American Power Symposium. 2017: pp. 1-6.
233. Hiremath RB, Shikha S, Ravindranath NH. Decentralized energy planning; modeling and application-a review. *Renewable Sustainable Energy Rev.* 2007;11(5):729-752.
234. Kanase-Patil AB, Saini RP, Sharma MP. Integrated renewable energy systems for off grid rural electrification of remote area. *Renewable Energy.* 2010;35(6):1342-1349.
235. Siddaiah R, Saini RP. A review on planning, configurations, modeling and optimization techniques of hybrid renewable energy systems for off grid applications. *Renewable Sustainable Energy Rev.* 2016;58:376-396.

236. Ashok S. Optimised model for community-based hybrid energy system. *Renewable Energy*. 2007;32(7):1155-1164.
237. Zeftawy AE, About El-Ela AA. Optimal planning of wind-diesel generation units in an isolated area. *Electr Pow Syst Res*. 1991;22:27-33.
238. Das TK, Chakraborty D, Seth S, Chak D, Bengal W. Energy consumption and prospects for renewable energy technologies in an Indian village. *Energy*. 1990;15(5):445-449.
239. Vose M, Liepins G. Punctuated equilibria in genetic search. *Complex Syst*. 1991;5:31-44.
240. Seyedmahmoudian M, Horan B, Soon TK, et al. State of the art artificial intelligence-based MPPT techniques for mitigating partial shading effects on PV systems—a review. *Renewable Sustainable Energy Rev*. 2016;64:435-455.
241. Silberholz J, Golden B. Comparison of metaheuristics. In: Gendreau M, Potvin J, eds. *Handbook of Metaheuristics*. Boston, MA: Springer; 2010:625-640.
242. Silva A, Marinheiro J, Cardoso HL, Oliveira E. Demand-side management in power grids: an ant colony optimization approach. Paper presented at: IEEE 18th International Conference on Computational Science and Engineering. 2015: pp. 300–306.
243. Yang G. Game theory-inspired evolutionary algorithm for global optimization. *Algorithms*. 2017;10:1-15.
244. Čepin M. *Assessment of Power System Reliability: Methods and Applications*. New York, NY: Springer; 2011.
245. Seyedmahmoudian M, Jamei E, Thirunavukkarasu G, et al. Short-term forecasting of the output power of a building-integrated photovoltaic system using a metaheuristic approach. *Energies*. 2018;11(5):1260.
246. Kim DH, Abraham A, Hirota K. Hybrid genetic particle swarm optimization algorithm. *Stud Comput Intell*. 2007;75: 147-170.

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